

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\E8301-AFinal.pvdb

Date of Analysis : Sep 9, 2026 10:22am

PV Elite 28, January 2026



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FileName : E8301-A_(Final)

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Vessel Design Summary:

Step: 25 10:22am Sep 9,2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 560.380 mm ID
Vessel Design Length, Tangent to Tangent 3366.09 mm
Distance of Bottom Tangent above Grade 0.00 mm
Distance of Base above Grade 0.00 mm
Specified Datum Line Distance 50.80 mm
Shell Side Design Temperature 230 °C
Channel Side Design Temperature 82 °C
Shell Side Design Pressure 350.000 KPa.
Channel Side Design Pressure 900.000 KPa.
Shop Shell Side Test Pressure 2537.813 KPa.
Field Shell Side Test Pressure 1635.376 KPa.
Shop Channel Side Test Pressure 2547.572 KPa.
Field Channel Side Test Pressure 1626.289 KPa.
Wind Design Code ASCE/SEI 7-16
Earthquake Design Code ASCE/SEI 7-16

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-106 B	...	...	K03006	
Head	SA-516 70	...	...	K02700	
Flange	SA-516 70	...	...	K02700	
Flange	SA-266 2	...	...	K03506	
Nozzle	SA-106 B	...	...	K03006	
Re-Pad	SA-516 70	...	...	K02700	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-213 TP316L	...	...	S31603	
Tubesheet	SA-266 2	...	...	K03506	
Flg Bolting	SA-193 B7	...	<= 64	G41400	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized

+Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (KPa. & mm):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Dish End	383.503	103.42	3710.000	3.2000	No	No
Main Shell	381.821	103.42	3208.000	3.2000	N/A	No
Inter Shell FL2	370.059	103.42	1258.000	3.2000	N/A	No
Inter Shell FL1	368.475	103.42	1260.000	3.2000	N/A	No
Main Shell	366.891	103.42	3223.000	3.2000	N/A	No
Shell Flg	360.029	103.42	1268.000	3.2000	N/A	No
Ch Shell	908.445	0.00	3232.000	3.2000	N/A	No
Ch Flg	901.584	0.00	1819.000	3.2000	N/A	No
Ch Cover	900.000	0.00	1820.000	3.2000	N/A	No

Liquid Level: 3417.95 mm Dens.: 0.001 kg./cm³ Sp. Gr.: 1.000



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Vessel Design Summary:

Step: 25 10:22am

Sep 9,2026

Element Types and Properties:

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	0.0	50.8	12.7	10.8	5.7	4.7	1.00	1.00
Cylinder	1200.0	1200.0	12.7	11.1	5.7	5.8	1.00	1.00
Body Flg	1351.0	147.8	10.0	44.5	...	...	1.00	1.00
Body Flg	1498.8	147.8	10.0	44.5	...	...	1.00	1.00
Cylinder	2198.8	700.0	12.7	11.1	5.7	5.2	1.00	1.00
Body Flg	2349.8	147.8	68.0	44.5	...	...	1.00	1.00
Cylinder	3119.8	700.0	12.7	11.1	5.7	...	1.00	1.00
Body Flg	3270.8	147.8	68.0	44.5	...	...	1.00	1.00
Body Flg	3315.3	44.4	65.0	44.5	...	...	1.00	1.00

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
N1-Tube Inlet	WRC-297	0.991	Nozzle	Passed
N2-Tube Outlet	WRC-297	0.991	Nozzle	Passed

Weights:

Fabricated - Bare w/o removable internals	2457.8 kg.
Shop Test - Fabricated + water (full)	3230.5 kg.
Shipping - Fab. + rem. intls.+ shipping app.	2457.8 kg.
Erected - Fab. + rem. intls.+ insul. (etc)	2457.8 kg.
Empty - Fab. + intls. + details + wghts.	2457.8 kg.
Operating - Empty + operating liquid (no ca)	3230.5 kg.
Field Test - Empty weight + test medium (full)	3230.5 kg.

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Nozzle Summary:

Step: 23 10:22am Sep 9,2026

Nozzle Calculation Summary (KPa. & mm):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
N1-Tube Inlet	1251	...	...	OK	5.70	OK	Passed
N2-Tube Outlet	1251	...	...	OK	5.70	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 1251 Nozzle : N2-Tube Outlet [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	Y Coordinate mm	Layout Angle deg	Dia. Limit mm
70	N1-Tube Inlet	2820.635	0.000	324.458
70	N2-Tube Outlet	2830.115	180.000	324.458

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.

lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Step: 22 10:22am Sep 9,2026

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
N1-Tube Inlet	150 mm	40	WNF	168.3	7.112	328.27	18.00	174	300
N2-Tube Outlet	150 mm	40	WNF	168.3	7.112	328.27	18.00	174	300

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N1-Tube Inl	SA-106 B	10.000	10.000	8.000	16.000	...
N2-Tube Out	SA-106 B	10.000	10.000	8.000	16.000	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm	Layout Angle deg	Proj Outside mm	Proj Inside mm	Installed in Component
N1-Tube Inlet	2769.835	0.0	150.00	0.00	Ch Shell
N2-Tube Outlet	2779.315	180.0	150.00	0.00	Ch Shell

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Bill of Material:

Step: 21 10:22am Sep 9,2026

Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	ELLIPTICAL HEAD: 2.0 X 1, 12.7mm THK X 560.4mm ID X 50.8mm	SA-516 70
1	CYLINDER: 12.7mm THK X 560.4mm ID X 1200.0mm	SA-106 B
2	WELD NECK FLANGE: 44.5mm THK X 749.3mm OD	SA-516 70
2	CYLINDER: 12.7mm THK X 560.4mm ID X 700.0mm	SA-106 B
2	WELD NECK FLANGE: 68.0mm THK X 749.3mm OD	SA-266 2
1	BLIND FLANGE: 65.0mm THK X 749.3mm OD	SA-266 2
1	TUBESHEET: 750.0mm X 70.0mm THK	SA-266 2
252	TUBES: 1830.0mm X 19.0mm DIA X 2.1mm THK	SA-213 TP316L
3	GASKET: 641.3mm OD X 536.6mm ID	...
60	BODY FLANGE BOLTS: 31.8mm DIA	SA-193 B7
100	NUTS FOR BODY FLANGE BOLTS: 31.8mm DIA	...
1	NAMEPLATE	...

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MDMT Summary:

Step: 24 10:22am Sep 9,2026

Minimum Design Metal Temperature Results Summary (°C):

Description	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm	E*	PWHT reqd
Notes								
Inter Shell F [5]	B	-18	-104		0.189		1.00	No
Bolting [21]		-48						
Inter Shell F [5]	B	-18	-104		0.188		1.00	No
Shell Flg [5]	B	-18	-104		0.184		1.00	No
Bolting [21]		-48						
Dish End [10]	B	-26	-104	-29	0.335	10.795	1.00	No
Dish End [7]	B	-21	-104	-29	0.273	12.700	1.00	No
Main Shell [8]	B	-26	-104	-29	0.198	10.795	1.00	No
Main Shell [8]	B	-26	-104	-29	0.191	11.113	1.00	No
Tubesheet: SS[13]	B	-11	-104	-29	0.170	17.500	1.00	No
Warmest MDMT:		-11	-104					
Ch Flg [5]	B	-18	-48		0.460		1.00	No
Ch Cover [5]	B	-18	-48		0.459		1.00	No
Ch Shell [8]	B	-26	-104	-29	0.189	11.113	1.00	No
N1-Tube Inlet [1]	B	-8	-104	-29	0.207	6.223	1.00	No
Nozzle Flg [5]	A	-18	-104					
N2-Tube Outlet[1]	B	-8	-104	-29	0.207	6.223	1.00	No
Nozzle Flg [5]	A	-18	-104					
Tubesheet: CS[14]	B	-11	-104	-29	0.170	17.500	1.00	No
Bolting [21]		-48						
Warmest MDMT:		-8	-48					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	-48.0	-28.9	Pass
Channel/Tube	-48.0	-28.9	Pass

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.

UCS-66(g) was not considered.

UCS-66(i) was not considered.



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MDMT Summary:

Step: 24 10:22am Sep 9, 2026

Notes:

Impact test temps were not entered in and not considered in the analysis.

UCS-66(i) applies to impact tested materials not by specification and

UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).

The Basic MDMT includes the (30F) PWHT credit if applicable.

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Warnings and Errors:

Step: 0 10:22am Sep 9, 2026

Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The level of precision is set to a value of less than 3. At this level you may not obtain proper resolution on some results. This value can be changed in Tools->Set Configuration Parameters->Default Values->Level of Precision.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Input Echo:

Step: 1 10:22am Sep 9,2026

Units used in this Analysis (MM):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	6.8946	KPa.
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm
Thickness	inches	25.4000	mm

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	350	KPa.
Channel Side Design Pressure	900.02	KPa.
Shell Side Design Temperature	230.0	°C
Channel Side Design Temperature	82.0	°C
Radiography, Shell Side	RT-2	
Radiography, Channel Side	RT-2	
Service Type, Shell Side	Air/Water/Steam	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-28.9	°C
MDMT (CET), Tube Side	-28.9	°C
User defined MAWP, Shell Side	0	KPa.
User defined MAWP, Channel Side	0	KPa.
User defined MAPnc, Shell Side	0	KPa.
User defined MAPnc, Channel Side	0	KPa.
User defined Test Pres., Shell Side	0	KPa.
User defined Test Pres., Channel Side	0	KPa.

Projection of Nozzle from Vessel Top	0	mm
Projection of Nozzle from Vessel Bottom	0	mm
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Vertical
Test Performed in Corroded Condition	No



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Input Echo:

Step: 1 10:22am Sep 9,2026

Field Pressure Test:

Type of Pressure Test UG-99(b)
 Pressure Test Position Vertical
 Test Performed in Corroded Condition Yes

Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-16
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed [V]	97.2 Km/hr
Surface Roughness Category	C: Open Terrain
Type of Surface	Moderately Smooth
Base Elevation	0 mm
Percent Wind for Hydrotest	50.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm
Distance Upwind of Crest Lh	0 mm
Distance from Crest to the Vessel x	0 mm
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm

Seismic Design Code	ASCE/SEI 7-16
Seismic Load Reduction Scale Factor	1.000
Importance Factor	1.250
Table Value Fa	1.300
Table Value Fv	1.500
Max. Mapped Res. Acceleration [Ss]	0.147
Max. Eff. Ground Acceleration [S]	0.065
Force Modification Factor R	2.000
Site Class	C
Component Elevation Ratio z/h	0.000
Amplification Factor Ap	1.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Moment Reduction Factor Tau	1.000

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y



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Input Echo:

Step: 1 10:22am Sep 9, 2026

Use ASME VIII-1 Appendix 1-9 N

Perform Blast Load Analysis No

Material Database Year Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk) Yes

Do not use Nozzle MDMT Interpretation VIII-1 01-37 No

Use Table G instead of exact equation for "A" Yes

Shell Head Joints are Tapered Yes

Compute "K" in corroded condition Yes

Use Code Case 2286 No

Use the MAWP to compute the MDMT Yes

For thickness ratios <= 0.35, MDMT will be -155F (-104C) Yes

For PWHT & P1 Materials the MDMT can be < -55F (-48C) No

Using Metric Material Databases, ASME II D Yes

Calculate B31.3 type stress for Nozzles with Loads Yes

Reduce the MDMT due to lower membrane stress Yes

Consider Longitudinal Stress in MDMT Calculations Yes

Complete Listing of Vessel Elements and Details:

Element#	1/9	
Element From Node	10	
Element To Node	20	
Element Type	Elliptical	
Description	Dish End	
Distance "FROM" to "TO"	50.8	mm
Inside Diameter	560.38	mm
Element Thickness	10.795	mm
Internal Corrosion Allowance	3.2	mm
Nominal Thickness	12.7	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	350	KPa.
Design Temperature Internal Pressure	230	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	230	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Allowable Stress, Ambient	138	N./mm ²
Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	220	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Efficiency, Longitudinal Seam	1.0	
Efficiency, Circumferential Seam	1.0	
Elliptical Head Factor	2.0	
Weld is pre-Heated	No	
Element From Node	10	
Detail Type	Liquid	
Detail ID	Liquid: 10	
Dist. from "FROM" Node / Offset dist	-120.75	mm
Height/Length of Liquid	171.55	mm
Liquid Density	0.0009984	kg./cm ³



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Input Echo:

Step: 1 10:22am Sep 9,2026

Element#	2/9
Element From Node	20
Element To Node	30
Element Type	Cylinder
Description	Main Shell
Distance "FROM" to "TO"	1200 mm
Inside Diameter	560.38 mm
Element Thickness	11.113 mm
Internal Corrosion Allowance	3.2 mm
Nominal Thickness	12.7 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	350 KPa.
Design Temperature Internal Pressure	230 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-106 B
Allowable Stress, Ambient	118 N./mm ²
Allowable Stress, Operating	118 N./mm ²
Allowable Stress, Hydrotest	216 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	202 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03006
Product Form	Smls. pipe
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	1200 mm
Liquid Density	0.0009984 kg./cm ³
Element From Node	20
Detail Type	Weight
Detail ID	flange weight
Dist. from "FROM" Node / Offset dist	800 mm
Miscellaneous Weight	250 Kg
Offset from Element Centerline	0 mm

Element#	3/9
Element From Node	30
Element To Node	40
Element Type	Flange
Description	Inter Shell FL2
Distance "FROM" to "TO"	147.83 mm
Flange Inside Diameter	536.58 mm
Element Thickness	44.45 mm
Internal Corrosion Allowance	3.2 mm
Nominal Thickness	10 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	350 KPa.
Design Temperature Internal Pressure	230 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 70
Allowable Stress, Ambient	138 N./mm ²



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Input Echo:

Step: 1 10:22am Sep 9, 2026

Allowable Stress, Operating	138	N./mm ²
Allowable Stress, Hydrotest	234	N./mm ²
Material Density	0.00775	kg./cm ³
P Number Thickness	32.004	mm
Yield Stress, Operating	220	N./mm ²
UCS-66 Chart Curve Designation	B	
External Pressure Chart Name	CS-2	
UNS Number	K02700	
Product Form	Plate	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	30	
Detail Type	Liquid	
Detail ID	Liquid: 30	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	161.6	mm
Liquid Density	0.0009984	kg./cm ³

Element#	4/9	
Element From Node	40	
Element To Node	50	
Element Type	Flange	
Description	Inter Shell FL1	
Distance "FROM" to "TO"	147.83	mm
Flange Inside Diameter	536.58	mm
Element Thickness	44.45	mm
Internal Corrosion Allowance	3.2	mm
Nominal Thickness	10	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	350	KPa.
Design Temperature Internal Pressure	230	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	130	°C
Effective Diameter Multiplier	1.2	
Material Name	SA-516 70	
Perform Flange Stress Calculation (Y/N)	Y	
Weld is pre-Heated	No	
Element From Node	40	
Detail Type	Liquid	
Detail ID	Liquid: 40	
Dist. from "FROM" Node / Offset dist	0	mm
Height/Length of Liquid	161.6	mm
Liquid Density	0.0009984	kg./cm ³

Element#	5/9	
Element From Node	50	
Element To Node	60	
Element Type	Cylinder	
Description	Main Shell	
Distance "FROM" to "TO"	700	mm
Inside Diameter	560.38	mm
Element Thickness	11.113	mm
Internal Corrosion Allowance	3.2	mm
Nominal Thickness	12.7	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	350	KPa.
Design Temperature Internal Pressure	230	°C
Design External Pressure	103.42	KPa.
Design Temperature External Pressure	130	°C



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Input Echo:

Step: 1 10:22am Sep 9,2026

Effective Diameter Multiplier	1.2
Material Name	SA-106 B
Allowable Stress, Ambient	118 N./mm ²
Allowable Stress, Operating	118 N./mm ²
Allowable Stress, Hydrotest	216 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	202 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03006
Product Form	Smls. pipe
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No
Element From Node	50
Detail Type	Liquid
Detail ID	Liquid: 50
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	700 mm
Liquid Density	0.0009984 kg./cm ³

Element#	6/9
Element From Node	60
Element To Node	70
Element Type	Flange
Description	Shell Flg
Distance "FROM" to "TO"	147.83 mm
Flange Inside Diameter	536.58 mm
Element Thickness	44.45 mm
Internal Corrosion Allowance	3.2 mm
Nominal Thickness	68 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	350 KPa.
Design Temperature Internal Pressure	230 °C
Design External Pressure	103.42 KPa.
Design Temperature External Pressure	130 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	136.8 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	208 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	60
Detail Type	Liquid
Detail ID	Liquid: 60
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	161.6 mm
Liquid Density	0.0009984 kg./cm ³



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Input Echo:

Step: 1 10:22am Sep 9,2026

Element From Node	70
Element To Node	80
Element Type	Cylinder
Description	Ch Shell
Distance "FROM" to "TO"	700 mm
Inside Diameter	560.38 mm
Element Thickness	11.113 mm
Internal Corrosion Allowance	3.2 mm
Nominal Thickness	12.7 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	900 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-106 B
Allowable Stress, Ambient	118 N./mm ²
Allowable Stress, Operating	118 N./mm ²
Allowable Stress, Hydrotest	216 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	223.6 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03006
Product Form	Smls. pipe
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	70
Detail Type	Liquid
Detail ID	Liquid: 70
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	700 mm
Liquid Density	0.0009984 kg./cm ³

Element From Node	70
Detail Type	Nozzle
Detail ID	N1-Tube Inlet
Dist. from "FROM" Node / Offset dist	350 mm
Nozzle Diameter	150 mm
Nozzle Schedule	40
Nozzle Class	300
Layout Angle	0.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	47.852 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B

Element From Node	70
Detail Type	Nozzle
Detail ID	N2-Tube Outlet
Dist. from "FROM" Node / Offset dist	359.48 mm
Nozzle Diameter	150 mm
Nozzle Schedule	40
Nozzle Class	300
Layout Angle	180.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	47.852 Kgf
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-106 B



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Input Echo:

Step: 1 10:22am Sep 9,2026

Element#	8/9
Element From Node	80
Element To Node	90
Element Type	Flange
Description	Ch Flg
Distance "FROM" to "TO"	147.83 mm
Flange Inside Diameter	536.58 mm
Element Thickness	44.45 mm
Internal Corrosion Allowance	3.2 mm
Nominal Thickness	68 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	900 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm
Yield Stress, Operating	230.08 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	80
Detail Type	Liquid
Detail ID	Liquid: 80
Dist. from "FROM" Node / Offset dist	0 mm
Height/Length of Liquid	161.6 mm
Liquid Density	0.0009984 kg./cm ³

Element#	9/9
Element From Node	90
Element To Node	100
Element Type	Flange
Description	Ch Cover
Distance "FROM" to "TO"	44.45 mm
Flange Outside Diameter	749.3 mm
Element Thickness	44.45 mm
Internal Corrosion Allowance	3.2 mm
Nominal Thickness	65 mm
External Corrosion Allowance	0 mm
Design Internal Pressure	900 KPa.
Design Temperature Internal Pressure	82 °C
Design External Pressure	0 KPa.
Design Temperature External Pressure	82 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No



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XY Coordinate Calculations:

Step: 2 10:22am

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XY Coordinate Calculations:

From	To	X (Horiz.) mm	Y (Vert.) mm	DX (Horiz.) mm	DY (Vert.) mm
Dish End		...	50.8	...	50.8
Main Shell		...	1250.8	...	1200
Inter Shell FL2		...	1401.8	...	147.828
Inter Shell FL1		...	1549.63	...	147.828
Main Shell		...	2249.63	...	700
Shell Flg		...	2400.63	...	147.828
Ch Shell		...	3170.63	...	700
Ch Flg		...	3321.64	...	147.828
Ch Cover		...	3366.09	...	44.45

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flg Calc [Int P]: Int Sh Fl 2

Flng: 21 10:22am

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Flange Input Data Values

Description: Int Sh Fl 2 :

Inter Shell FL2

Flange Type	Integral Weld Neck		
Design Pressure	P	370.06	KPa.
Design Temperature		230	°C
Internal Corrosion Allowance	ci	3.2000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	536.575	mm
Flange Outside Diameter	A	749.300	mm
Flange Thickness	t	44.4500	mm
Thickness of Hub at Small End	go	11.1125	mm
Thickness of Hub at Large End	gl	36.5125	mm
Length of Hub	h	103.3780	mm
Flange Material		SA-516 70	
Flange Material UNS number		K02700	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	692.150	mm
Nominal Bolt Diameter	a	31.7500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		20	
Flange Face Outside Diameter	Fod	641.350	mm
Flange Face Inside Diameter	Fid	536.575	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	641.350	mm
Gasket Inside Diameter	Gi	536.575	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Axial Force on Flange		1760.00	Kgf
Bending Moment on Flange		130.00	Kg-m.
Flange Class		150	
Flange Grade		GR 1.1	



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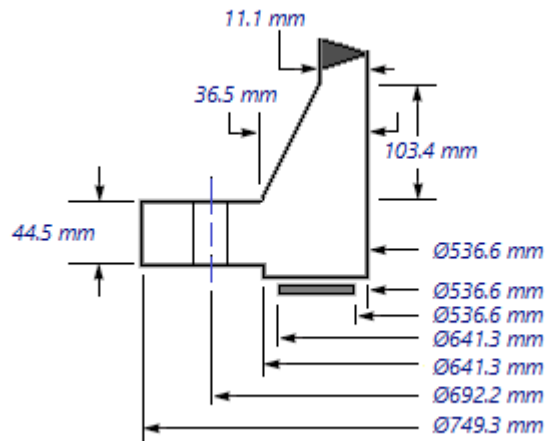
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Flg Calc [Int P]: Int Sh Fl 2

Flng: 21 10:22am

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (370.06 \cdot (536.575/2 + 3.2)) / (118.0 \cdot 1.0 - 0.6 \cdot 370.06) + C_a$$

$$= 4.0531 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 7.9125) / (271.4875 + 0.6 \cdot 7.9125)$$

$$= 3379.745 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 11.1125) / (268.2875 + 0.6 \cdot 11.1125)$$

$$= 4768.690 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	41.250 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	542.975 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	33.312 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	7.912 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	41.275 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	49.188 mm
Basic Gasket Width, $b_o = N/2$	24.594 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{b_o}$	12.497 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	616.356 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 616.3564^2 \cdot 370.059$$

$$= 11259.551 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 12.4968 \cdot \pi \cdot 616.3564 \cdot 3.0 \cdot 370.06$$

$$= 5478.979 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 542.975^2 \cdot 370.0589$$

$$= 8738.099 \text{ Kgf}$$



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Flg Calc [Int P]: Int Sh Fl 2

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Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 11260 - 8738$$

$$= 2521.452 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(11260 + 5479 + 0, 0)$$

$$= 16738.529 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 12.4968 \cdot \pi \cdot 616.356 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 170132.234 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((16739 + 1760 + 4 \cdot 130 / 616.36) / 172, 170132 / 172)$$

$$= 97.005 \text{ cm}^2$$

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 119.871 \cdot 172.0 / (68.95 \cdot \pi (641.35 + 536.58))$$

$$= 8.080 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172.0 \cdot 119.8707$$

$$= 210235.03 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 16739 - 11260$$

$$= 5478.98 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (692.15 - 616.3564) / 2$$

$$= 37.8968 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + h_g)$$

$$= 0.5 ((692.15 - 542.975) / 2 + 37.8968)$$

$$= 56.2422 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g_1) / 2$$

$$= (692.15 - 542.975 - 33.312) / 2$$

$$= 57.9313 \text{ mm}$$

The following values have been computed from Table 4.16.7:

Determine Geometry Factors [Aa, Bb, Cc, Ddg]:

$$A_a = 103.162, B_b = 41.25, C_c = 103.378 \text{ and } D_{dg} = 20.613 \text{ mm}$$

Compute the value of [Kab]:

$$= A_a \cdot B_b^3 \cdot (1/3 - 0.210 \cdot (B_b / A_a) \cdot (1 - 1/12 \cdot (B_b / A_a)^4))$$

$$= 103.162 \cdot 41.25^3 \cdot (1/3 - 0.210 \cdot (41.25 / 103.162) \cdot (1 - 1/12 \cdot (41.25 / 103.162)^4))$$

$$= 180.692 \text{ cm}^4$$

Compute the value of [Kcd]:

$$= C_c \cdot D_{dg}^3 \cdot (1/3 - 0.105 \cdot (D_{dg} / C_c) \cdot (1 - 1/192 \cdot (D_{dg} / C_c)^4))$$

$$= 103.38 \cdot 20.613^3 \cdot (1/3 - 0.105 \cdot (20.613 / 103.38) \cdot (1 - 1/192 \cdot (20.613 / 103.38)^4))$$



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Flg Calc [Int P]: Int Sh Fl 2

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= 28.283 cm⁴

Flange Inertia Term [Ip,term1]:

= Kab + Kcd = 110.265 + 17.259

= 208.975 cm⁴

Flange Inertia Term [Ip,term2]:

= $Ar \cdot t^3 \left(\frac{1}{3} - 0.21 \left(\frac{t}{AR} \right) \left(1 - \frac{1}{12} \left(\frac{t}{AR} \right)^4 \right) \right)$

= $103.162 \cdot 41.25^3 \left(\frac{1}{3} - 0.21 \left(\frac{41.25}{103.162} \right) \left(1 - \frac{1}{12} \left(\frac{41.25}{103.162} \right)^4 \right) \right)$

= 180.692 cm⁴

Flange Inertia [Ip]:

= max(Ip,term1, Ip,term2)

= max(209.0, 180.7)

= 209.0 cm⁴

Flange Moment of Inertia for Integral Types [I]:

= $0.0874 \cdot L \cdot go^2 \cdot ho \cdot B / V$

= $0.0874 \cdot 0.851 \cdot 7.912^2 \cdot 65.546 \cdot 542.975 / 0.031$

= 526.232 cm⁴

Summary of Moments for Internal Pressure: (Kg-m.)

Loading		Force	Distance	Bolt Corr	Moment
End Pressure,	Md	8738.	57.9313	1.0167	515.
Face Pressure,	Mt	2521.	56.2422	1.0167	144.
Gasket Load,	Mg	5479.	37.8968	1.0167	211.
Gasket Seating,	Matm	210235.	37.8968	1.0167	8100.
Moment Load,	Moe	...	...	...	147.

Moment Value, eq. 4.16.16, [Moe]:

= $4Me \left[\frac{1}{(0.384 \cdot IP + I)} \right] \cdot \left(\frac{hD}{(C-2 \cdot hD)} \right) + FA \cdot hD$

= $4 \cdot 130 \left[\frac{526.2}{(0.384 \cdot 209.0 + 526.2)} \right] \cdot \left(\frac{57.9}{(692.2 - 2 \cdot 57.9)} \right) +$

$1760 \cdot 57.931$

= 147.318 Kg-m.

Total Moment for Operation, Mo

1017. Kg-m.

Total Moment for Gasket seating, Mg

8100. Kg-m.

You chose not to perform Stress Calculations on this Standard Flange.

The pressure rating of the flange will be used to check code compliance.

Estimated Finished Weight of Flange at given Thk. 107.6 kg.

Estimated Unfinished Weight of Forging at given Thk 246.1 kg.

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction

-104 °C

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Flg Calc [Int P]: Int Sh Fl 1

Flng: 22 10:22am

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Flange Input Data Values

Description: Int Sh Fl 1 :

Inter Shell FL1

Flange Type	Integral Weld Neck		
Design Pressure	P	368.47	KPa.
Design Temperature		230	°C
Internal Corrosion Allowance	ci	3.2000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	536.575	mm
Flange Outside Diameter	A	749.300	mm
Flange Thickness	t	44.4500	mm
Thickness of Hub at Small End	go	11.1125	mm
Thickness of Hub at Large End	gl	36.5125	mm
Length of Hub	h	103.3780	mm
Flange Material		SA-516 70	
Flange Material UNS number		K02700	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	692.150	mm
Nominal Bolt Diameter	a	31.7500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		20	
Flange Face Outside Diameter	Fod	641.350	mm
Flange Face Inside Diameter	Fid	536.575	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	641.350	mm
Gasket Inside Diameter	Gi	536.575	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	



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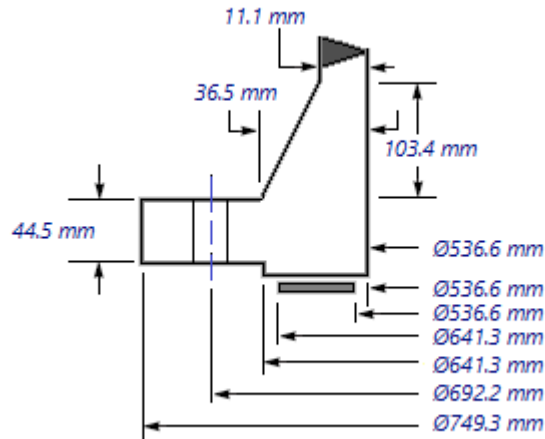
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Flg Calc [Int P]: Int Sh Fl 1

Flng: 22 10:22am

Sep 9, 2026



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (368.47 \cdot (536.575/2 + 3.2)) / (118.0 \cdot 1.0 - 0.6 \cdot 368.47) + C_a$$

$$= 4.0494 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 7.9125) / (271.4875 + 0.6 \cdot 7.9125)$$

$$= 3379.745 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 11.1125) / (268.2875 + 0.6 \cdot 11.1125)$$

$$= 4768.690 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	41.250 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	542.975 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	33.312 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	7.912 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	41.275 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	49.188 mm
Basic Gasket Width, $bo = N/2$	24.594 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{bo}$	12.497 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	616.356 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 616.3564^2 \cdot 368.475$$

$$= 11211.354 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 12.4968 \cdot \pi \cdot 616.3564 \cdot 3.0 \cdot 368.47$$

$$= 5455.526 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 542.975^2 \cdot 368.4749$$

$$= 8700.695 \text{ Kgf}$$



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Flg Calc [Int P]: Int Sh Fl 1

Flng: 22 10:22am

Sep 9, 2026

Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 11211 - 8701$$

$$= 2510.659 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(11211 + 5456 + 0, 0)$$

$$= 16666.879 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 12.4968 \cdot \pi \cdot 616.356 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 170132.234 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((16667 + 0 + 4 \cdot 0 / 616.36) / 172, 170132 / 172)$$

$$= 97.005 \text{ cm}^2$$

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 119.871 \cdot 172.0 / (68.95 \cdot \pi (641.35 + 536.58))$$

$$= 8.080 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172.0 \cdot 119.8707$$

$$= 210235.03 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 16667 - 11211$$

$$= 5455.52 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (692.15 - 616.3564) / 2$$

$$= 37.8968 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + h_g)$$

$$= 0.5 ((692.15 - 542.975) / 2 + 37.8968)$$

$$= 56.2422 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g_1) / 2$$

$$= (692.15 - 542.975 - 33.312) / 2$$

$$= 57.9313 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	8701.	57.9313	1.0167	512.
Face Pressure, Mt	2511.	56.2422	1.0167	144.
Gasket Load, Mg	5456.	37.8968	1.0167	210.
Gasket Seating, Matm	210235.	37.8968	1.0167	8100.

Total Moment for Operation, Mo

866. Kg-m.

Total Moment for Gasket seating, Mg

8100. Kg-m.

You chose not to perform Stress Calculations on this Standard Flange.



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Flg Calc [Int P]: Int Sh Fl 1

Flng: 22 10:22am

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The pressure rating of the flange will be used to check code compliance.

Estimated Finished Weight of Flange at given Thk. 107.6 kg.

Estimated Unfinished Weight of Forging at given Thk 246.1 kg.

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -104 °C

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Flg Calc [Int P]: Shell Flg

Flng: 23 10:22am Sep 9,2026

Flange Input Data Values

Description: Shell Flg :

Shell Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	360.03	KPa.
Design Temperature		230	°C
Internal Corrosion Allowance	ci	3.2000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	536.575	mm
Flange Outside Diameter	A	749.300	mm
Flange Thickness	t	44.4500	mm
Thickness of Hub at Small End	go	11.1125	mm
Thickness of Hub at Large End	gl	36.5125	mm
Length of Hub	h	103.3780	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	136.80	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	692.150	mm
Nominal Bolt Diameter	a	31.7500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		20	
Flange Face Outside Diameter	Fod	641.350	mm
Flange Face Inside Diameter	Fid	536.575	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	641.350	mm
Gasket Inside Diameter	Gi	536.575	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	



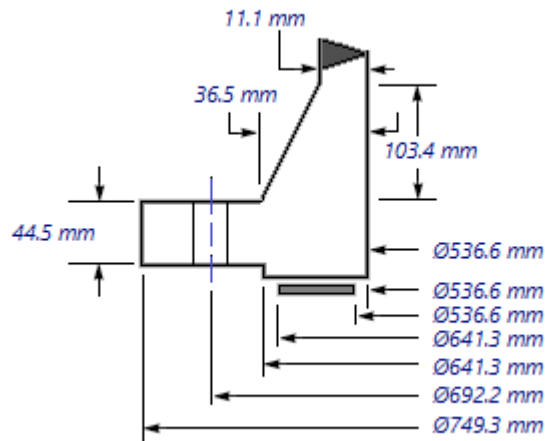
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Flg Calc [Int P]: Shell Flg

Flng: 23 10:22am Sep 9, 2026



ASME Code, Section VIII Division 1, 2025

Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (360.03 \cdot (536.575/2 + 3.2)) / (118.0 \cdot 1.0 - 0.6 \cdot 360.03) + C_a$$

$$= 4.0299 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 7.9125) / (271.4875 + 0.6 \cdot 7.9125)$$

$$= 3379.745 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 11.1125) / (268.2875 + 0.6 \cdot 11.1125)$$

$$= 4768.690 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	41.250 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	542.975 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	33.312 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	7.912 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	41.275 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	49.188 mm
Basic Gasket Width, $bo = N/2$	24.594 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{bo}$	12.497 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	616.356 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 616.3564^2 \cdot 360.029$$

$$= 10954.390 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 12.4968 \cdot \pi \cdot 616.3564 \cdot 3.0 \cdot 360.03$$

$$= 5330.485 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 542.975^2 \cdot 360.0294$$

$$= 8501.276 \text{ Kgf}$$



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Flg Calc [Int P]: Shell Flg

Flng: 23 10:22am

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Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 10954 - 8501$$

$$= 2453.114 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(10954 + 5330 + 0, 0)$$

$$= 16284.876 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 12.4968 \cdot \pi \cdot 616.356 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 170132.234 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((16285 + 0 + 4 \cdot 0 / 616.36) / 172, 170132 / 172)$$

$$= 97.005 \text{ cm}^2$$

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 119.871 \cdot 172.0 / (68.95 \cdot \pi (641.35 + 536.58))$$

$$= 8.080 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172.0 \cdot 119.8707$$

$$= 210235.03 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 16285 - 10954$$

$$= 5330.49 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (692.15 - 616.3564) / 2$$

$$= 37.8968 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + h_g)$$

$$= 0.5 ((692.15 - 542.975) / 2 + 37.8968)$$

$$= 56.2422 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g_1) / 2$$

$$= (692.15 - 542.975 - 33.312) / 2$$

$$= 57.9313 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	8501.	57.9313	1.0167	501.
Face Pressure, Mt	2453.	56.2422	1.0167	140.
Gasket Load, Mg	5330.	37.8968	1.0167	205.
Gasket Seating, Matm	210235.	37.8968	1.0167	8100.

Total Moment for Operation, Mo

846. Kg-m.

Total Moment for Gasket seating, Mg

8100. Kg-m.

You chose not to perform Stress Calculations on this Standard Flange.



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Flg Calc [Int P]: Shell Flg

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The pressure rating of the flange will be used to check code compliance.

Estimated Finished Weight of Flange at given Thk. 107.6 kg.

Estimated Unfinished Weight of Forging at given Thk 246.1 kg.

Flange Stress results per WRC-538 at computed Sfmax (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	475.43	475.43	1.000	475.43	499.99	0.951
Tangential Flange	St	158.90	356.57	0.446	158.90	374.99	0.424
Tangential Flange	Sto	207.66	237.71	0.874	207.66	250.00	0.831
Combined	Sh/f + St	634.33	713.14	0.889	634.33	749.99	0.846
Combined	Sr + St	617.90	713.14	0.866	617.90	749.99	0.824
Combined	Sh + Sto	683.09	713.14	0.958	683.09	749.99	0.911
Radial Flange	Sr	459.00	475.43	0.965	459.00	499.99	0.918
Maximum Stress Ratio				1.000			0.951

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 0.9746/5.5828)

= 1.0000

Hub Stress Correction Factor [f2]:



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Flg Calc [Int P]: Shell Flg

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= max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)
 = max(0.25; 6.0823/7.0203) if h/ho < 0.35, else max (0.25; 0.8967/2.0529)
 = 0.4368

In this case, the hub ratio h/ho, 1.5772 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	536.5 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	5.9935 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	208.0 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.243 deg

Gasket Contact Area [Ag]:

= $\pi/4 (GO.D.^2 - GI.D.^2)$
 = $\pi/4 (641.35^2 - 536.575^2)$
 = 969.3 cm²

Target Bolt Stress [Sbsel]: (Eq. O-1)

= $SgT \cdot Ag / (nb \cdot Abs)$
 = $207.0 \cdot 969.316 / (20 \cdot 5.994)$
 = 1673.9 N./mm²

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

= min(Sbsel; Sbmax)
 = min(1673.873; 507.5)
 = 507.5 N./mm²

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

= max(Sbsel; Sbmin)
 = max(507.5; 290.0)
 = 507.5 N./mm²

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

= min(Sbsel; Sfmax)
 = min(507.5; 536.498)
 = 507.5 N./mm²

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

:: Sbsel $\geq$ Sgmin-S(Ag / (nb • Abs))
 :: 507.5 $\geq$ 140.0(969.316 / (20 • 5.994))
 :: 507.5 $\geq$ 1132.1 N./mm² Check Fails

Check if Gasket Operating Stress is maintained. (Eq. O-8)

:: Sbsel $\geq$ (Sgmin-O • Ag + $\pi/4$ • Pmax • GI.D.²) / (ϕG • Abs • nb)
 :: 507.5 $\geq$ (97.0 • 969.316 + $\pi/4$ • 360.029 • 536.575²) / (0.7 • 5.994 • 20)
 :: 507.5 $\geq$ 1130.2 N./mm² Check Fails



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Flg Calc [Int P]: Shell Flg

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Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: S_{bsel} \leq S_{gmax} (A_g / (n_b \cdot Abs))$$
$$:: 507.5 \leq 276.0 (969.316 / (20 \cdot 5.994))$$
$$:: 507.5 \leq 2231.8 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: S_{bsel} \leq S_{fmax} (\theta_{tagmax} / \theta_{tafmax})$$
$$:: 507.5 \leq 536.498 (1.0 / 0.243)$$
$$:: 507.5 \leq 2205.2 \text{ N./mm}^2 \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= S_{bsel} \cdot K \cdot Abs \cdot \phi_b$$
$$= 507.5 \cdot 0.2 \cdot 5.994 \cdot 31.75$$
$$= 197.0 \text{ Kg-m.}$$
Bolt Allowable Stress for: SA-193 B8 Cl. 2 SbB8 112 N./mm²Bolt Allowable Stress for: SA-193 B7 Sb 171 N./mm²

Note: Since $S_b \geq S_{bB8}$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction

-104 °C

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Flg Calc [Int P]: New Flange

Flng: 24 10:22am Sep 9,2026

Flange Input Data Values

Description: New Flange :

Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	901.58	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.2000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	536.575	mm
Flange Outside Diameter	A	749.300	mm
Flange Thickness	t	44.4500	mm
Thickness of Hub at Small End	go	11.1125	mm
Thickness of Hub at Large End	gl	36.5125	mm
Length of Hub	h	103.3780	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	692.150	mm
Nominal Bolt Diameter	a	31.7500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		20	
Flange Face Outside Diameter	Fod	641.350	mm
Flange Face Inside Diameter	Fid	536.575	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	641.350	mm
Gasket Inside Diameter	Gi	536.575	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	



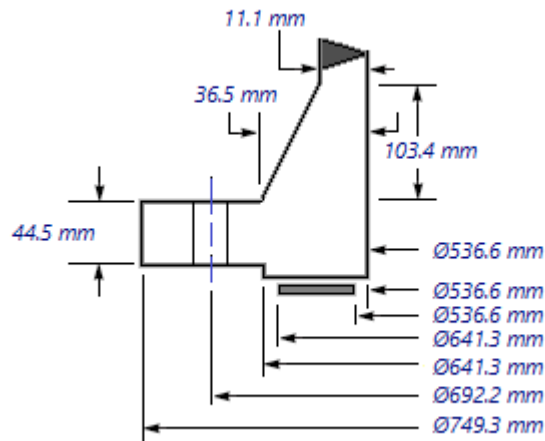
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Flg Calc [Int P]: New Flange

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + C_a)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (901.58 \cdot (536.575/2 + 3.2)) / (118.0 \cdot 1.0 - 0.6 \cdot 901.58) + C_a$$

$$= 5.2840 \text{ mm}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 7.9125) / (271.4875 + 0.6 \cdot 7.9125)$$

$$= 3379.745 \text{ KPa.}$$

Hub Small End Hub MAPnc:

$$= (S_a \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 11.1125) / (268.2875 + 0.6 \cdot 11.1125)$$

$$= 4768.690 \text{ KPa.}$$

Corroded Flange Thickness, $t_c = T - c_i$	41.250 mm
Corroded Flange ID, $B_{cor} = B + 2 \cdot c_i$	542.975 mm
Corroded Large Hub, $g1_{cor} = g1 - c_i$	33.312 mm
Corroded Small Hub, $g0_{cor} = g0 - c_i$	7.912 mm
Code R Dimension, $R = ((C - B_{cor})/2) - g1_{cor}$	41.275 mm

Gasket Contact Width, $N = (G_o - G_i)/2$	49.188 mm
Basic Gasket Width, $bo = N/2$	24.594 mm
Effective Gasket Width, $b = C_b \cdot \sqrt{bo}$	12.497 mm
Gasket Reaction Diameter, $G = G_o - 2 \cdot b$	616.356 mm

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 616.3564^2 \cdot 901.584$$

$$= 27431.932 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 12.4968 \cdot \pi \cdot 616.3564 \cdot 3.0 \cdot 901.58$$

$$= 13348.577 \text{ Kgf}$$

Hydrostatic End Load at Flange ID [Hd]:

$$= \pi/4 \cdot B_{cor}^2 \cdot P$$

$$= \pi/4 \cdot 542.975^2 \cdot 901.584$$

$$= 21288.854 \text{ Kgf}$$



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Flg Calc [Int P]: New Flange

Flng: 24 10:22am

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Pressure Force on Flange Face [Ht]:

$$= H - H_d$$

$$= 27432 - 21289$$

$$= 6143.078 \text{ Kg}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(W_{o,1} + W_{o,2} + H'p, 0)$$

$$= \max(27432 + 13349 + 0, 0)$$

$$= 40780.512 \text{ Kg}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 12.4968 \cdot \pi \cdot 616.356 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 170132.234 \text{ Kg}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4Me/G) / S_b, W_g / S_a)$$

$$= \max((40781 + 0 + 4 \cdot 0 / 616.36) / 172, 170132 / 172)$$

$$= 97.005 \text{ cm}^2$$

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 119.871 \cdot 172.0 / (68.95 \cdot \pi (641.35 + 536.58))$$

$$= 8.080 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172.0 \cdot 119.8707$$

$$= 210235.03 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 40781 - 27432$$

$$= 13348.58 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (692.15 - 616.3564) / 2$$

$$= 37.8968 \text{ mm}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + h_g)$$

$$= 0.5 ((692.15 - 542.975) / 2 + 37.8968)$$

$$= 56.2422 \text{ mm}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g_1) / 2$$

$$= (692.15 - 542.975 - 33.312) / 2$$

$$= 57.9313 \text{ mm}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	21289.	57.9313	1.0167	1254.
Face Pressure, Mt	6143.	56.2422	1.0167	351.
Gasket Load, Mg	13349.	37.8968	1.0167	514.
Gasket Seating, Matm	210235.	37.8968	1.0167	8100.

Total Moment for Operation, Mo 2119. Kg-m.

Total Moment for Gasket seating, Mg 8100. Kg-m.

You chose not to perform Stress Calculations on this Standard Flange.



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Flg Calc [Int P]: New Flange

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The pressure rating of the flange will be used to check code compliance.

Estimated Finished Weight of Flange at given Thk. 107.6 kg.

Estimated Unfinished Weight of Forging at given Thk 246.1 kg.

Flange Stress results per WRC-538 at computed S_{fmax} (N./mm²):

		Actual	Operating Allowed	stress ratio	Gasket Actual	Seating Allowed	stress ratio
Longitudinal Hub	Sh	499.99	499.99	1.000	499.99	499.99	1.000
Tangential Flange	St	167.11	374.99	0.446	167.11	374.99	0.446
Tangential Flange	Sto	218.39	250.00	0.874	218.39	250.00	0.874
Combined	Sh/f + St	667.10	749.99	0.889	667.10	749.99	0.889
Combined	Sr + St	649.83	749.99	0.866	649.83	749.99	0.866
Combined	Sh + Sto	718.38	749.99	0.958	718.38	749.99	0.958
Radial Flange	Sr	482.72	499.99	0.965	482.72	499.99	0.965
Maximum Stress Ratio				1.000			1.000

Coefficients for f and f2

	<i>f</i>	<i>f2</i>	<i>f3</i>
<i>a</i>	-0.71375912	-0.01447638	-0.06456312
<i>b</i>	-0.12846279	-0.27814745	-0.08232239
<i>c</i>	1.08037907	0.01035395	0.13691677
<i>d</i>	0.99766848	1.37984158	-0.77879888
<i>e</i>	1.21823466	-0.34346764	0.14909019
<i>f</i>	0.01024612	0.02536192	0.00660062
<i>g</i>	0.4262313	-0.0014508	0.0616252
<i>h</i>	1.4204976	2.25543162	0.74639834
<i>i</i>	-0.70278181	-0.60889597	-0.079595
<i>j</i>	-0.02483937	-0.15735462	0.09851673
<i>k</i>	-1.59460436	1.22516062	-0.1121436

$$f_n = \frac{\left[a + c \left(\frac{g1}{g0} \right) + e \left(\frac{h}{h0} \right) + g \left(\frac{g1}{g0} \right)^2 + i \left(\frac{h}{h0} \right)^2 + k \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}{\left[1 + b \left(\frac{g1}{g0} \right) + d \left(\frac{h}{h0} \right) + f \left(\frac{g1}{g0} \right)^2 + h \left(\frac{h}{h0} \right)^2 + j \left(\frac{g1}{g0} \right) \left(\frac{h}{h0} \right) \right]}$$

Hub Stress Correction Factor [f]:

= max(1; F1num / F1den)

= max(1; 0.9746/5.5828)

= 1.0000

Hub Stress Correction Factor [f2]:



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Flg Calc [Int P]: New Flange

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= max(0.25; fn2) if h/ho < 0.35, else max (0.25; fn3)
 = max(0.25; 6.0823/7.0203) if h/ho < 0.35, else max (0.25; 0.8967/2.0529)
 = 0.4368

In this case, the hub ratio h/ho, 1.5772 was ≥ 0.35 .

PV Elite iterates on the assembly bolt load stress, Sfmax, until it exceeds the limits provided by WRC 538.

Bolting and Gasket Calculations per ASME PCC-1 Appendix O:

Input Data:

Maximum permissible bolt stress	Sbmax	507.5 N./mm ²
Minimum permissible bolt stress	Sbmin	290.0 N./mm ²
Max. permissible bolt stress prior to flange damage	Sfmax	564.2 N./mm ²
Target assembly gasket stress	SgT	207.0 N./mm ²
Maximum permissible gasket stress	Sgmax	276.0 N./mm ²
Minimum gasket operating stress	Sgmin-O	97.0 N./mm ²
Minimum gasket seating stress	Sgmin-S	140.0 N./mm ²
Nut Factor	K	0.200
Fraction of Gasket Load remaining after relaxation	Phig	0.700
Root Area of a Single Bolt	Abs	5.9935 cm ²
Max. permissible single flange rotation for gasket	Thetagmax	1.000 deg

Calculated Results:

Flange Material Ambient Yield Stress	Sya	250.0 N./mm ²
Flange Material Operating Yield Stress	Syo	230.1 N./mm ²
Single flange rotation at Sfmax	Thetafmax	0.256 deg

Gasket Contact Area [Ag]:

= $\pi/4 (GO.D.^2 - GI.D.^2)$
 = $\pi/4 (641.35^2 - 536.575^2)$
 = 969.3 cm²

Target Bolt Stress [Sbsel]: (Eq. O-1)

= $SgT \cdot Ag / (nb \cdot Abs)$
 = $207.0 \cdot 969.316 / (20 \cdot 5.994)$
 = 1673.9 N./mm²

Upper Bolt Stress Limit [Sbsel]: (Eq. O-4)

= min(Sbsel; Sbmax)
 = min(1673.873; 507.5)
 = 507.5 N./mm²

Lower Bolt Stress Limit [Sbsel]: (Eq. O-5)

= max(Sbsel; Sbmin)
 = max(507.5; 290.0)
 = 507.5 N./mm²

Upper Bolt Stress Limit including the Flange [Sbsel]: (Eq. O-6)

= min(Sbsel; Sfmax)
 = min(507.5; 564.215)
 = 507.5 N./mm²

Check if Gasket Assembly Seating Stress is achieved. (Eq. O-7)

:: Sbsel $\geq$ Sgmin-S(Ag / (nb • Abs))
 :: 507.5 $\geq$ 140.0(969.316 / (20 • 5.994))
 :: 507.5 $\geq$ 1132.1 N./mm² Check Fails

Check if Gasket Operating Stress is maintained. (Eq. O-8)

:: Sbsel $\geq$ (Sgmin-O • Ag + $\pi/4$ • Pmax • GI.D.²) / (ϕG • Abs • nb)
 :: 507.5 $\geq$ (97.0 • 969.316 + $\pi/4$ • 901.584 • 536.575²) / (0.7 • 5.994 • 20)
 :: 507.5 $\geq$ 1144.8 N./mm² Check Fails



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Flg Calc [Int P]: New Flange

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Check if Gasket Maximum Stress is achieved. (Eq. O-9)

$$:: S_{bsel} \leq S_{gmax}(A_g / (n_b \cdot A_{bs}))$$
$$:: 507.5 \leq 276.0(969.316 / (20 \cdot 5.994))$$
$$:: 507.5 \leq 2231.8 \text{ N./mm}^2 \text{ Check Passes}$$

Check if Flange Rotation Limit is exceeded. (Eq. O-10)

$$:: S_{bsel} \leq S_{fmax}(\theta_{tagmax} / \theta_{tafmax})$$
$$:: 507.5 \leq 564.215(1.0/0.256)$$
$$:: 507.5 \leq 2205.2 \text{ N./mm}^2 \text{ Check Passes}$$

Assembly Bolt Torque [Tb]: (Eq. O-2)

$$= S_{bsel} \cdot K \cdot A_{bs} \cdot \phi_b$$
$$= 507.5 \cdot 0.2 \cdot 5.994 \cdot 31.75$$
$$= 197.0 \text{ Kg-m.}$$
Bolt Allowable Stress for: SA-193 B8 Cl. 2 $S_{bB8} \ 118 \text{ N./mm}^2$ Bolt Allowable Stress for: SA-193 B7 $S_b \ 171 \text{ N./mm}^2$

Note: Since $S_b \geq S_{bB8}$, bolt allowable stress comparison check Passes (see Div 1 UG-44(b)).

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction

-48 °C

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Flg Calc [Int P]: New Flange

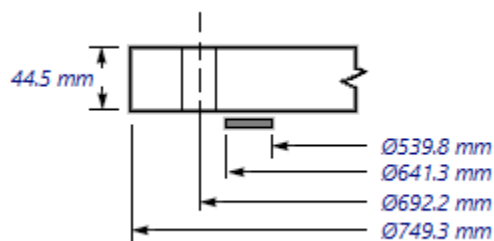
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Flange Input Data Values

Description: New Flange :

Ch Cover

Flange Type		Blind	
Design Pressure	P	900.00	KPa.
Design Temperature		82	°C
Internal Corrosion Allowance	ci	3.2000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	749.300	mm
Flange Thickness	t	44.4500	mm
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Perimeter along the Center of the Bolts	L	2174.453	mm
Diameter of Bolt Circle	C	692.150	mm
Nominal Bolt Diameter	a	31.7500	mm
Type of Threads		TEMA Thread Series	
Number of Bolts		20	
Flange Face Outside Diameter	Fod	641.350	mm
Flange Face Inside Diameter	Fid	539.750	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	641.350	mm
Gasket Inside Diameter	Gi	539.750	mm
Gasket Factor	m	3.0000	
Gasket Design Seating Stress	y	68.95	N./mm ²
Gasket Thickness	tg	3.1750	mm
Raised face depth	Fd	0.0000	mm
Flange Class		150	
Flange Grade		GR 1.1	



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Gasket Contact Width,	$N = (Go - Gi)/2$	50.800	mm
Basic Gasket Width,	$bo = N/2$	25.400	mm
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	12.700	mm
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	615.950	mm



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Flg Calc [Int P]: New Flange

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Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:

$$= \pi/4 \cdot G^2 \cdot P$$

$$= \pi/4 \cdot 615.95^2 \cdot 900.0$$

$$= 27347.639 \text{ Kgf}$$

Contact Load on Gasket Surfaces [Hp or Wo,2]:

$$= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$$

$$= 2 \cdot 12.7 \cdot \pi \cdot 615.95 \cdot 3.0 \cdot 900.0$$

$$= 13532.853 \text{ Kgf}$$

Design Bolt Load for the Operating Condition [Wo]:

$$= \max(Wo,1 + Wo,2 + H'p, 0)$$

$$= \max(27348 + 13533 + 0, 0)$$

$$= 40880.492 \text{ Kgf}$$

Design Bolt Load for the Gasket Seating Condition [Wg]:

$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 68.95 \cdot 12.7 \cdot \pi \cdot 615.95 + 0.0 \cdot 0.0 \cdot 0.0$$

$$= 172784.422 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((Wo + Fa + 4Me/G) / Sb, Wg/Sa)$$

$$= \max((40880 + 0 + 4 \cdot 0 / 615.95) / 172, 172784 / 172)$$

$$= 98.517 \text{ cm}^2$$

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= Ab \cdot Sa / (y \cdot \pi (Go + Gi))$$

$$= 119.871 \cdot 172.0 / (68.95 \cdot \pi (641.35 + 539.75))$$

$$= 8.059 \text{ mm}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= Sa \cdot Ab$$

$$= 172.0 \cdot 119.8707$$

$$= 210235.03 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= Wo$$

$$= 40880.49 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (692.15 - 615.95) / 2$$

$$= 38.1000 \text{ mm}$$

*You chose not to perform Stress Calculations on this Standard Flange.
The pressure rating of the flange will be used to check code compliance.*

Estimated Finished Weight of Flange at given Thk.	151.9 kg.
Estimated Unfinished Weight of Forging at given Thk	151.9 kg.

Flange MDMT including temperature reduction per 3.11:

Flange MDMT with Temp reduction -48 °C

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Internal Pressure Calculations: Step: 8 10:22am Sep 9,2026

Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd KPa.	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) N./mm ²	Element#
Dish End		383.5	12.7	3.2	560.38	138	1
Main Shell		381.82	12.7	3.2	560.38	118	2
Inter Shell FL2		370.06	10	3.2	536.58	138	3
Inter Shell FL1		368.47	10	3.2	536.58	138	4
Main Shell		366.89	12.7	3.2	560.38	118	5
Shell Flg		360.03	68	3.2	536.58	136.8	6
Ch Shell		908.45	12.7	3.2	560.38	118	7
Ch Flg		901.58	68	3.2	536.58	138	8
Ch Cover		900	65	3.2	749.3	138	9

Element Required Thickness and MAWP :

From	To	Design Pressure KPa.	M.A.W.P. Corroded KPa.	M.A.P. New & Cold KPa.	Minimum Thickness mm	Required Thickness mm	Element#
Dish End		350	3710.08	5295.98	10.795	5.7	1
Main Shell		350	3208.31	4570.82	11.1125	5.7	2
Inter Shell FL2		350	1257.98	1960.01	44.45	No Calc	3
Inter Shell FL1		350	1259.57	1960.01	44.45	No Calc	4
Main Shell		350	3223.24	4570.82	11.1125	5.7	5
Shell Flg		350	1268.01	1960.01	44.45	No Calc	6
Ch Shell		900	3231.69	4570.82	11.1125	5.7	7
Ch Flg		900	1818.85	1960.01	44.45	No Calc	8
Ch Cover		900	1820.43	1960.01	44.45	No Calc	9

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 1258 KPa.
 Shell Side MAPnc = 1960 KPa.
 Channel Side MAWP = 1251 KPa.
 Channel Side MAPnc = 1960 KPa.

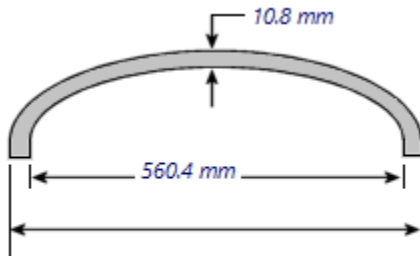
Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

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Internal Pressure Results for: Dish End

Elliptical Head Design Information:					
Design Pressure	P	383.503 KPa.	Int. Design Temperature	230 °C	
Material		SA-516 70	Ext. Design Temperature	230 °C	
Length		50.800 mm	External Pressure Chart	CS-2	
			UNS Number	K02700	
Int. Corr. All.	c	3.2000 mm	Allowable Stress (ope)	S	138.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	138.0 N./mm ²
Inside Diameter	Di	560.380 mm	Actual Stress	14.1 N./mm ²	
Spec. Min. Thk t,e, or sa		10.7950 mm	Spec. Nominal Thk.	12.700 mm	
Surface Area		4654.4 cm ²	SG of Contents	1.000	
Element Volume		35570.3 cm ³	Weight of Contents	34.9 Kg	
Empty Weight		47.7 Kg	Operating Weight	82.6 Kg	
			Aspect Ratio	2.00	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 7.005 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.502 mm

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (383.503 \cdot 566.78 \cdot 0.985) / (2 \cdot 138.0 \cdot 1.0 - 0.2 \cdot 383.503)$$

$$= 0.7762 + 3.2000 = 3.9762 \text{ mm}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 2.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 33.503 KPa.

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 7.595) / (0.985 \cdot 566.78 + 0.2 \cdot 7.595)$$

$$= 3744 - 34 = 3710 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 138.0 \cdot 1.0 \cdot 10.795) / (1.0 \cdot 560.38 + 0.2 \cdot 10.795)$$



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Element: Dish End

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= 5296 KPa.

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (383.503 \cdot (0.985 \cdot 566.78 + 0.2 \cdot 7.595)) / (2 \cdot 1.0 \cdot 7.595)$$

$$= 14.137 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (383.503 \cdot 283.39) / (138.0 \cdot 1.0 - 0.6 \cdot 383.503) + 3.2$$

$$= 3.989 \text{ mm}$$

Straight Flange Maximum Allowable Working Pressure:

[Less Operating Hydrostatic Head Pressure of 33.503 KPa.](#)

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (138.0 \cdot 1.0 \cdot 9.5) / (283.39 + 0.6 \cdot 9.5)$$

$$= 4535 - 34 = 4501 \text{ KPa.}$$

Factor K, corroded condition [Kcor]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth})))^2 / 6$$

$$= (2 + (566.78 / (2 \cdot 143.295)))^2 / 6$$

$$= 0.985195$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 9.374 %

[Note: Please Check Requirements of UCS-79 as Elongation is > 5%.](#)

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 10.795, tr = 2.548, c = 3.2 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.335$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 12.7, tr = 2.598, c = 3.2 mm, E* = 1.0

Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.273$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-21 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1257.981 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	10.795 mm	Basic MDMT	-26 °C
Stress/Thk. Ratio	0.335	Temperature Difference	77 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

[Note: Temperature values displayed above are for the location with the highest basic MDMT](#)

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 7.005 mm

Head inner surface maximum deviation inside the specified shape, 0.625% of D: 3.502 mm

EXTERNAL PRESSURE RESULTS for: Dish End



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Element: Dish End

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Elliptical Head From 10 to 20 Ext. Chart: CS-2 at 230 °C

Elastic Modulus from Chart: CS-2 at 230 °C: 189879744 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
7.595	581.97	76.63	0.0018126	84.04

$$MAEP = B / (K0 * Do / t) = 84.0365 / (0.9 * 76.6254) = 1218.5056 \text{ KPa.}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
1.546	581.97	376.38	0.0003690	35.04

$$MAEP = B / (K0 * Do / t) = 35.0363 / (0.9 * 376.377) = 103.4255 \text{ KPa.}$$

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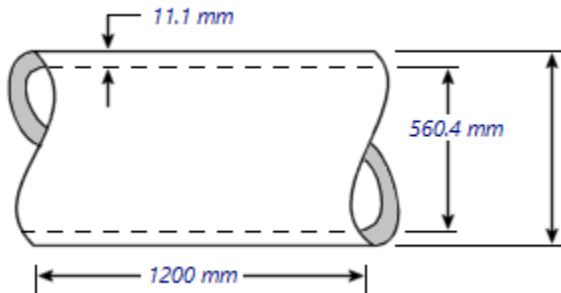
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Element: Main Shell

Step: 10 10:22am Sep 9,2026

Internal Pressure Calculation Results:

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Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:			
Design Pressure P	381.821 KPa.	Int. Design Temperature	230 °C
Material	SA-106 B	Ext. Design Temperature	130 °C
Length	1200.000 mm	External Pressure Chart	CS-2
		UNS Number	K03006
Int. Corr. All. c	3.2000 mm	Allowable Stress (ope) S	118.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	118.0 N./mm ²
Inside Diameter Di	560.380 mm	Actual Stress	13.9 N./mm ²
Spec. Min. Thk t,e, or sa	11.1125 mm	Spec. Nominal Thk.	12.700 mm
Surface Area	22083.4 cm ²	SG of Contents	1.000
Element Volume	202087.8 cm ³	Weight of Contents	202.0 Kg
Empty Weight	212.7 Kg	Operating Weight	414.6 Kg

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (381.821 \cdot 283.39) / (118.0 \cdot 1.0 - 0.6 \cdot 381.821)$$

$$= 0.9188 + 3.2000 = 4.1188 \text{ mm}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 2.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 31.821 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 7.9125) / (283.39 + 0.6 \cdot 7.9125)$$

$$= 3240 - 32 = 3208 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 11.1125) / (280.19 + 0.6 \cdot 11.1125)$$

$$= 4571 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (381.821 \cdot (283.39 + 0.6 \cdot 7.9125)) / (1.0 \cdot 7.9125)$$



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Element: Main Shell

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= 13.905 N./mm²

Minimum Design Metal Temperature Results:

Govrn. thk, $t_g = 10.795$, $t_r = 1.507$, $c = 3.2$ mm, $E^* = 1.0$
 Thickness Ratio = $t_r * E^* / (t_g - c) = 0.198$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1257.981 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	10.795 mm	Basic MDMT	-26 °C
Stress/Thk. Ratio	0.198	Temperature Difference	77 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E^*	100 %

EXTERNAL PRESSURE RESULTS for: Main Shell

Cylindrical Shell From 20 to 30 Ext. Chart: CS-2 at 130 °C

Elastic Modulus from Chart: CS-2 at 130 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
7.912	582.60	1297.50	73.63	2.2271	0.0009183	82.64

$MAEP = (4*B) / (3*(Do/t)) = (4*82.6364) / (3*73.631) = 1496.3196$ KPa.

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
2.603	582.60	1297.50	223.80	2.2271	0.0001737	17.36

$MAEP = (4*B) / (3*(Do/t)) = (4*17.3612) / (3*223.8001) = 103.4268$ KPa.

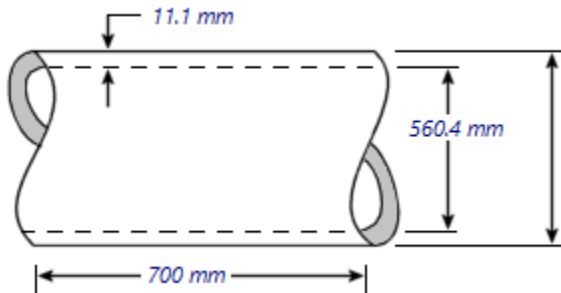
Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
7.912	582.60	37094.20	73.63	50.0000	0.0002089	20.88

$MAEP = (4*B) / (3*(Do/t)) = (4*20.8832) / (3*73.631) = 378.1375$ KPa.



Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:			
Design Pressure P	366.891 KPa.	Int. Design Temperature	230 °C
Material	SA-106 B	Ext. Design Temperature	130 °C
Length	700.000 mm	External Pressure Chart	CS-2
		UNS Number	K03006
Int. Corr. All. c	3.2000 mm	Allowable Stress (ope) S	118.0 N./mm ²
Ext. Corr. All. ce	0.0000 mm	Allowable Stress (amb) Sa	118.0 N./mm ²
Inside Diameter Di	560.380 mm	Actual Stress	13.4 N./mm ²
Spec. Min. Thk t,e, or sa	11.1125 mm	Spec. Nominal Thk.	12.700 mm
Surface Area	12882.0 cm ²	SG of Contents	1.000
Element Volume	122388.6 cm ³	Weight of Contents	122.3 Kg
Empty Weight	124.0 Kg	Operating Weight	246.4 Kg

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (366.891 \cdot 283.39) / (118.0 \cdot 1.0 - 0.6 \cdot 366.891)$$

$$= 0.8828 + 3.2000 = 4.0828 \text{ mm}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 2.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 16.891 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 7.9125) / (283.39 + 0.6 \cdot 7.9125)$$

$$= 3240 - 17 = 3223 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 11.1125) / (280.19 + 0.6 \cdot 11.1125)$$

$$= 4571 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (366.891 \cdot (283.39 + 0.6 \cdot 7.9125)) / (1.0 \cdot 7.9125)$$



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 Element: Main Shell Step: 11 10:22am Sep 9,2026

= 13.361 N./mm²

Minimum Design Metal Temperature Results:

Govrn. thk, tg = 11.113, tr = 1.507, c = 3.2 mm, E* = 1.0
 Thickness Ratio = tr * E*/(tg - c) = 0.191, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -26 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1257.981 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	11.113 mm	Basic MDMT	-26 °C
Stress/Thk. Ratio	0.191	Temperature Difference	78 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

EXTERNAL PRESSURE RESULTS for: Main Shell

Cylindrical Shell From 50 to 60 Ext. Chart: CS-2 at 130 °C

Elastic Modulus from Chart: CS-2 at 130 °C: 199943392 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
7.912	582.60	700.00	73.63	1.2015	0.0017823	101.08

MAEP = (4*B)/(3*(Do/t)) = (4*101.0833)/(3*73.631) = 1830.342 KPa.

Results for Required Thickness				Tca		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
2.018	582.60	700.00	288.65	1.2015	0.0002240	22.39

MAEP = (4*B)/(3*(Do/t)) = (4*22.3912)/(3*288.6456) = 103.425 KPa.

Results for Maximum Stiffened Length				Slen		
Tca	Outer Dia	Slen	Do/t	L/D	Factor A	Factor B
7.912	582.60	14626.36	73.63	25.1051	0.0002100	21.00

MAEP = (4*B)/(3*(Do/t)) = (4*20.9987)/(3*73.631) = 380.23 KPa.

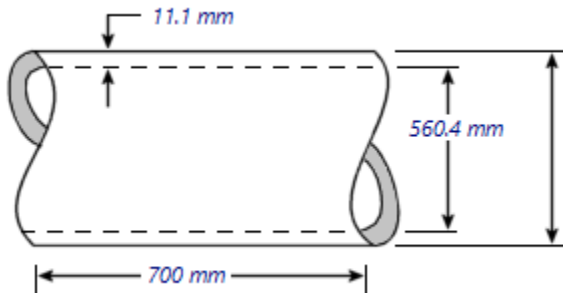


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 Element: Ch Shell

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025



Internal Pressure Results for: Ch Shell

Cylindrical Shell Design Information:					
Design Pressure	P	908.445 KPa.	Int. Design Temperature	82 °C	
Material		SA-106 B	Ext. Design Temperature	82 °C	
Length		700.000 mm	External Pressure Chart	CS-2	
			UNS Number	K03006	
Int. Corr. All.	c	3.2000 mm	Allowable Stress (ope)	S	118.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa	118.0 N./mm ²
Inside Diameter	Di	560.380 mm	Actual Stress	33.1 N./mm ²	
Spec. Min. Thk t,e, or sa		11.1125 mm	Spec. Nominal Thk.	12.700 mm	
Surface Area		12882.0 cm ²	SG of Contents	1.000	
Element Volume		172675.6 cm ³	Weight of Contents	172.6 Kg	
Empty Weight		124.0 Kg	Operating Weight	296.6 Kg	

Radiography Information:			
User Defined		User Defined	
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (908.445 \cdot 283.39) / (118.0 \cdot 1.0 - 0.6 \cdot 908.445)$$

$$= 2.1920 + 3.2000 = 5.3920 \text{ mm}$$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 2.5000 mm per UG-16 will be used.

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 8.445 KPa.

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 7.9125) / (283.39 + 0.6 \cdot 7.9125)$$

$$= 3240 - 8 = 3232 \text{ KPa.}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118.0 \cdot 1.0 \cdot 11.1125) / (280.19 + 0.6 \cdot 11.1125)$$

$$= 4571 \text{ KPa.}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (908.445 \cdot (283.39 + 0.6 \cdot 7.9125)) / (1.0 \cdot 7.9125)$$



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Element: Ch Shell

Step: 12 10:22am

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= 33.083 N./mm²

Minimum Design Metal Temperature Results:

Govern. thk, tg = 11.113, tr = 1.499, c = 3.2 mm, E* = 1.0

Thickness Ratio = $tr * E^* / (tg - c) = 0.189$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -26 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

MDMT Design Information:			
MDMT Curve	B	Press. at MDMT	1250.991 KPa.
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-29 °C
Governing Thk.	11.113 mm	Basic MDMT	-26 °C
Stress/Thk. Ratio	0.189	Temperature Difference	78 °C
UCS-66(c) red.	No	Computed Min. Temp.	-104 °C
		MDMT per UG-20(f)	-29 °C
		Joint Efficiency E*	100 %

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Shop Pressure Test Results:

Step: 13 10:22am Sep 9, 2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1635.376 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	455.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2537.813 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1383.780 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	500.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	500.500 KPa.
App 27-4	= M.A.W.P.	1257.981 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1626.289 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1170.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2547.572 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1376.091 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1287.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1287.000 KPa.
App 27-4	= M.A.W.P.	1250.991 KPa.

Vertical Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Dish End	67.0	234.0	0.286	2571.40
Main Shell	66.3	216.0	0.307	2569.53
Main Shell	66.0	216.0	0.305	2554.87
Ch Shell	66.0	216.0	0.306	2556.32

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube Inlet	Nozzle	118.00	118.00	1.000
N1-Tube Inlet	Pad	138.00	138.00	1.000
N2-Tube Outlet	Nozzle	118.00	118.00	1.000
N2-Tube Outlet	Pad	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Dish End	138.00	138.00	1.000
Main Shell	118.00	118.00	1.000
Inter Shell FL2	138.00	138.00	1.000
Inter Shell FL1	138.00	138.00	1.000
Main Shell	118.00	118.00	1.000
Shell Flg	138.00	136.80	1.009
Ch Shell	118.00	118.00	1.000
Ch Flg	138.00	138.00	1.000
Ch Cover	138.00	138.00	1.000

Minimum

1.000



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Shop Pressure Test Results:

Step: 13 10:22am

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Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	115.00	115.00	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube Inlet	33.50	216.00	0.155
N2-Tube Outlet	33.50	216.00	0.155

Elements Suitable for Test Pressure.

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Field Pressure Test Results: Step: 14 10:22am Sep 9, 2026

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1635.376 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	455.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2537.813 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1383.780 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	500.500 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	500.500 KPa.
App 27-4	= M.A.W.P.	1257.981 KPa.

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	1626.289 KPa.
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	1170.000 KPa.
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	2547.572 KPa.
UG100	= 1.10 * M.A.W.P. * Sa/S	1376.091 KPa.
PED	= max(1.43*DP, 1.25*DP*ratio)	1287.000 KPa.
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	1287.000 KPa.
App 27-4	= M.A.W.P.	1250.991 KPa.

Vertical Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & KPa.):

From To	Stress	Allowable	Ratio	Pressure
Dish End	61.5	234.0	0.263	1668.96
Main Shell	60.7	216.0	0.281	1667.09
Main Shell	60.2	216.0	0.279	1652.43
Ch Shell	59.5	216.0	0.276	1635.03

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
N1-Tube Inlet	Nozzle	118.00	118.00	1.000
N1-Tube Inlet	Pad	138.00	138.00	1.000
N2-Tube Outlet	Nozzle	118.00	118.00	1.000
N2-Tube Outlet	Pad	138.00	138.00	1.000

Minimum

1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Dish End	138.00	138.00	1.000
Main Shell	118.00	118.00	1.000
Inter Shell FL2	138.00	138.00	1.000
Inter Shell FL1	138.00	138.00	1.000
Main Shell	118.00	118.00	1.000
Shell Flg	138.00	136.80	1.009
Ch Shell	118.00	118.00	1.000
Ch Flg	138.00	138.00	1.000
Ch Cover	138.00	138.00	1.000

Minimum

1.000



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Field Pressure Test Results:

Step: 14

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Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	115.00	115.00	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum

1.000

Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
N1-Tube Inlet	21.41	216.00	0.099
N2-Tube Outlet	21.41	216.00	0.099

Elements Suitable for Test Pressure.

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ASME TS Calc: Pressure Test

Case: 1 10:22a Sep 9, 2026

Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Test Pressure		1635.38 KPa.
Shell Thickness	ts	12.7000 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	560.380 mm
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K03006
Shell Allowable Stress at Temperature	Ss	118.00 N./mm ²
Shell Allowable Stress at Ambient		118.00 N./mm ²

Channel Data:

Channel Description: Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		1626.29 KPa.
Channel Thickness	tc	12.7000 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	560.380 mm
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K03006
Channel Allowable Stress at Temperature	Sc	118.00 N./mm ²
Channel Allowable Stress at Ambient		118.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	252
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	1830.00 mm
Straight Tube Length (bet. inner tubsht faces) L		1760.00 mm
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-213 TP316L
Tube Material UNS Number		S31603
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		115.00 N./mm ²
Tube Allowable Stress At Ambient		115.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	170.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	28.5750 mm
Tube Layout Pattern		Triangular

Fillet Weld Leg	af	1.6510 mm
Groove Weld Leg	ag	1.6510 mm
Tube-Tubesheet Joint Weld Type		Seal/No Weld
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No

Radius to Outermost Tube Hole Center	ro	222.201 mm
Largest Center-to-Center Tube Distance	U1	0.0000 mm
Length of Expanded Portion of Tube	ltx	46.8000 mm
Tube-side pass partition groove depth	hg	7.0000 mm
Area of Tube Layout	Ap	1127.2 cm ²
Perimeter of Tube Layout	Cp	1307.9 mm



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ASME TS Calc: Pressure Test

Case: 1 10:22a Sep 9, 2026

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral with Channel, Conf. e

Tubesheet Design Metal Temperature T 21.11 °C
Tubesheet Material SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03506
Tubesheet Allowable Stress at Temperature S 138.00 N./mm²
Tubesheet Allowable Stress at Ambient Tt 138.00 N./mm²
Thickness of Tubesheet h 70.0000 mm
Tubesheet Corr. Allowance (Shell side) Cats 0.0000 mm
Tubesheet Corr. Allowance (Channel side) Catc 0.0000 mm
Tubesheet Outside Diameter A 750.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped? No
Area of the Untubed Lanes AL 177.2 cm²
Junction Stress Reduction option Perform Plastic Calculation
Perform Differential Pressure Design No
Run Multiple Load Cases Yes
Shell Side Min. Design Pressure Psd,min 103.4190 KPa.

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side Shell
Flange Outside Diameter A 750.000 mm
Flange Inside Diameter B 536.575 mm
Flange Face Outside Diameter Fod 641.350 mm
Flange Face Inside Diameter Fid 536.575 mm
Gasket Outside Diameter Go 641.350 mm
Gasket Inside Diameter Gi 536.575 mm
Small end Hub thk. g0 11.1125 mm
Large end Hub thk. g1 36.5125 mm
Gasket Factor, m 3.00
Gasket Design Seating Stress y 68.95 N./mm²
Flange Facing Sketch Code Sketch 1a
Gasket Thickness tg 3.1750 mm
Full face Gasket Flange Option Program Selects

Bolting Information:

Diameter of Bolt Circle C 692.150 mm
Nominal Bolt Diameter dB 31.7500 mm
Thread Series TEMA Thread Series
Number of Bolts n 20
Bolt Material SA-193 B7
Bolt Allowable Stress At Temperature Sb 172.00 N./mm²
Bolt Allowable Stress At Ambient Sa 172.00 N./mm²
Weld between Flange and Shell/Channel 0.0000 mm

Note: The full bolt load $Sa \cdot Ab$, was used for all load cases.

Tubesheet Integral with Channel
Tubesheet Extended as Flange Yes
Thickness of Extended Portion of Tubesheet Tf 54.0000 mm
Is Bolt Load Transferred to the Tubesheet Yes



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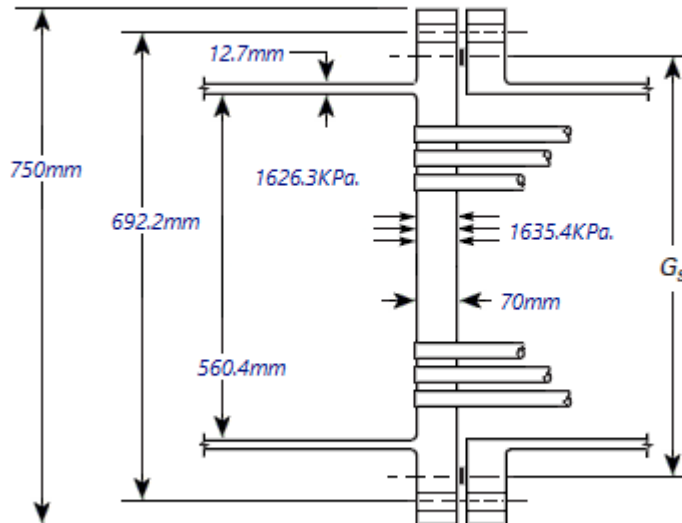
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ASME TS Calc: Pressure Test

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Tubesheet Gasketed w/Shell and Integral w/Channel, Extended as a Flange



Configuration e:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TM-1 Material Group G

Elastic Mod. at Tubesheet Design Temp.	21.1 °C	0.19512E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
= $(P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
= $(1729.71 \cdot (19.05/2 - 0)) / (115.0 \cdot 1.0 + 0.4 \cdot 1729.71)$
= $0.1424 + 0.0000 = 0.1424$ mm

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-4 at 21.11 °C
Elastic Modulus for Material 193048800.00 KPa.



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ASME TS Calc: Pressure Test

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Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	1982.20	9.03	50.0000	0.0134948	99.11

EMAWP = $(2.167 / (D/T) - 0.0833) * B = 15530.9688 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.4306	19.05	1982.20	44.24	50.0000	0.0005621	54.26

EMAWP = $(4 * B) / (3 * (D/T)) = (4 * 54.2626) / (3 * 44.2361) = 1635.4484 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.4306 mm
Allowable Internal Pressure at Corroded thickness	27949.72 KPa.
Required Internal Design Pressure	1729.71 KPa.
Allowable External Pressure at Corroded thickness	15530.97 KPa.
Required External Design Pressure	1635.38 KPa.
Required Thickness due to Shell Side pressure	0.4306 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	52.388 mm
Basic Gasket Width,	$b0 = N / 2$	26.194 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) * 2.5$	12.892 mm
Gasket Reaction Diameter,	$G = Go - 2.0 * b$	615.566 mm

Results for ASME U-tube Tubesheet Calculations for Configuration e:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$= 2 * ro + dt$
 $= 2 * 222.201 + 19.05$
 $= 463.452 \text{ mm}$

Basic ligament efficiency for shear [μ]:

$= (p - dt) / p$
 $= (28.575 - 19.05) / 28.575$
 $= 0.333$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$= Gs / Do$ (Configurations d, e, f)
 $= 615.5656 / 463.452$
 $= 1.3282$

Ratio [pc]:

$= Dc / Do$ (Configurations a, e, f)
 $= 560.38 / 463.452$
 $= 1.2091$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$= ltx / h$
 $= 46.8 / 70.0$



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ASME TS Calc: Pressure Test

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$$= 0.669 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$\begin{aligned} &= \max(dt - 2tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2tt) \\ &= \max(19.05 - 2 \cdot 2.11 (0.19512E+09/.20270E+09) \cdot \\ &\quad (114/137) \cdot (0.669); 19.05 - 2 \cdot 2.11) \\ &= 16.7868 \text{ mm} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(A1; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) } \\ &= 28.575 / \sqrt{ (1 - 4 \cdot \min(177.16 ; 4 \cdot 463.452 \cdot 28.575) / (\pi \cdot 463.452^2)) } \\ &= 30.2050 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (30.205 - 16.7868) / 30.205 \\ &= 0.4442 \end{aligned}$$

$$h/p = 2.449694 ; \mu^* = 0.444237 ; E^* = 95575416 \text{ KPa.}$$

$$E^*/E = 0.471509 ; \nu^* = 0.303610$$

Note: As h/p (2.450) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

β_s , k_s , λ_s , δ_s and ω_s are all = 0

Channel coefficient [β_c]:

$$\begin{aligned} &= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{ ((Dc + tc) tc) } \\ &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((560.38 + 12.7) 12.7) } \\ &= 0.0213 \text{ 1/mm} \end{aligned}$$

Channel coefficient [k_c]:

$$\begin{aligned} &= \beta_c \cdot E_c \cdot tc^3 / (6(1 - \nu_s^2)) \\ &= 0.021 \cdot 0.20270E+09 \cdot 12.7^3 / (6(1 - 0.3^2)) \\ &= 1620395648.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$\begin{aligned} &= 6(Dc + tc/\cos(\theta)) \cdot k_c/h^3 \cdot (CmTheta + \\ &\quad h \cdot \beta_c \cdot (Cm\Delta + CqTheta)/2 + (h \cdot \beta_c)^2/2 \cdot Cq\Delta) \\ &= 6(560.38 + 12.7/\cos(0)) \cdot 0.16204E+1/70.0^3 \cdot (1.0 + \\ &\quad 70.0 \cdot 0.021 \cdot (1.0 + 1.0)/2 + (70.0 \cdot 0.021)^2/2 \cdot 1.0) \\ &= 58542868.0 \text{ KPa.} \end{aligned}$$

Channel coefficient [δ_c]:

$$\begin{aligned} &= Dc^2 / (4 \cdot E_c \cdot tc) \cdot (1 - Dc \cdot \nu_c / (2(Dc + tc))) \\ &= 560.38^2 / (4 \cdot 0.20270E+09 \cdot 12.7) \cdot (1 - 560.38 \cdot 0.3 / (2(560.38 + 12.7))) \\ &= 0.26022E-01 \text{ mm/N./mm}^2 \end{aligned}$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ω_c]:

$$\begin{aligned} &= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \\ &\quad \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)) \\ &= 1.209 \cdot 0.16204E+1 [0.26022 (1.0 + 0.021 \cdot 1.0 \cdot 70.0) - \\ &\quad 0 (1.0/70.0 + 1.0 \cdot 0.021/2)) \\ &= 2706.9341 \text{ mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:



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ASME TS Calc: Pressure Test

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Diameter ratio [K]:

$$= A / D_o$$

$$= 750.0/463.452$$

$$= 1.6183$$

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda c + E \cdot \ln(K))$$

$$= (1 - 0.3) / 95575416 \cdot (58542868 + 0.20270E+09 \cdot \ln(1.62))$$

$$= 1.1375$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2/16 \cdot [(\psi_s - 1) \cdot (\psi_s^2 + 1) \cdot P_s - (\psi_c - 1) \cdot (\psi_c^2 + 1) \cdot P_t]$$

$$= 463.452^2/16 \cdot [(1.328 - 1) \cdot (1.328^2 + 1) \cdot 1635.376 - (1.209 - 1) \cdot (1.209^2 + 1) \cdot 1626.289]$$

$$= 8675971.0000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]

$$= M_{ts} + (\omega_c \cdot P_t) + W^* \cdot (C - G_s) / (2 \cdot \pi \cdot D_o)$$

$$= 8675971 + (4402256) + 210235 \cdot (692.15 - 615.566) / (2 \cdot \pi \cdot 463.452)$$

$$= 67298840.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^*) - D_o^2/32 \cdot F(P_s - P_t) / (1 + F)$$

$$= (67298840) - 463.452^2/32 \cdot 1.138 (1635.38 - 1626.29) / (1 + 1.14)$$

$$= 31452142.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2/64 \cdot (3 + \nu^*) \cdot (P_s - P_t)$$

$$= 31452142 + 463.452^2/64 \cdot (3 + 0.304) \cdot (1635.38 - 1626.29)$$

$$= 31552892.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|M_p|; |M_o|)$$

$$= \max(|31452142|; |31552892|)$$

$$= 31552892.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 31552892 / (0.4442 (70.0 - 7.0)^2)$$

$$= 107.3790 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot \beta_T \cdot S_y$$

$$= 2 \cdot 0.95 \cdot 250.0$$

$$= 474.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 12320578 / (0.4814 (25.0133 - 7.0)^2)$$

$$= 473.2969 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 25.0133 + 0 + 0$$

$$= 25.0133 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(25.0133; 0.0124)$$



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ASME TS Calc: Pressure Test

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= 25.0133 mm

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [$|P_s - P_t|$]:

= $|P_s - P_t|$

= $|1635.38 - 1626.29|$

= 9.087 KPa.

Shear Stress check [τ_{limit}]:

= $4 \cdot \mu \cdot h/Do \cdot 0.8 \cdot \beta T \cdot S_y$

= $4 \cdot 0.333 \cdot 70.0/463.45 \cdot 0.8 \cdot 0.95 \cdot 249.995$

= 38.263 N./mm²

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

= $1/(4 \cdot \mu \cdot (0.8 \cdot \beta T \cdot S_y)) \cdot (4 \cdot A_p/C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$

= $1/(4 \cdot 0.333 \cdot 189.997) \cdot (4 \cdot 1127.2/1307.93) \cdot |1635.38 - 1626.29| + 0$

= 0.0124 mm

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Channel thickness of: 12.700 mm

Min. Channel len. adjacent to the tubesheet is: 151.850mm

Tube Metal Area [A_t]:

= $\pi(d_o - t)t$

= $\pi(19.05 - 2.11)2.11$

= 1.123 cm²

Stress/Force summary for loadcase D3 un-corr. ($P_{sd,max} + P_{td,max}$):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	107.4 <=	475.0 N./mm ²	Ok
Tubesheet shear stress	0.0 <=	190.0 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. ($P_{sd,max} + P_{td,max}$):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	25.013	70.000	Ok
Tube-Tubesheet Fillet Weld Leg :	0.000	1.651	Ok
Tube-Tubesheet Groove Weld Leg :	0.000	1.651	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

	--Reqd. Thk. + CA		---- Tubesheet		Stresses		Case	Pass/
Case#	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd	Type	Fail
D1uc	24.129	32.011	57	475	6	190	Ps+Pt D1	Ok
D2uc	37.328	...	160	475	6	190	Ps+Pt D2	Ok
D3uc	25.013	...	107	475	...	190	Ps+Pt D3	Ok
Max:	37.3281	32.011 mm		0.337		0.034 (Str. Ratio)		

Load Case Definitions:

[P_s & P_t]:



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ASME TS Calc: Pressure Test

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Shell-side and Tube-side Design or Operating Pressures
derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
Load Case Tables

[c]:
With or Without Corrosion Allowance

[D1, D2, D3]:
Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

-- Integral Cyl. Req'd. Thk.	Shell	Stress	Channel	Stress	Pass
Case#	Shell	Channel	Stress	Allwd	Fail
D1uc	...	...	...	...	Ok
D2uc	...	...	...	...	Ok
D3uc	...	...	...	...	Ok
Max	...	...	...	...	

Note:
Because there was no net pressure differential across the tubesheet due to
the specified pressures psd,min and ptd,min; load case D4 was not processed as
the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	37.328	70.000	Ok
Tubesheet thickness flanged extension :	32.011	54.000	Ok
Tube thickness :	0.431	2.110	Ok
Tube-tubesheet fillet weld leg :	0.000	1.651	Ok
Tube-tubesheet groove weld leg :	0.000	1.651	Ok

Given Shell Thickness: 12.7000 mm

Given Channel Thickness: 12.7000 mm

Min Channel length of thk, (12.700) adj. to tubesheet: 151.850 mm

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External Pressure Calculations: Step: 15 10:22am Sep 9,2026

External Pressure Calculation Results Summary:

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External Pressure Calculations:

From	To	Section Length mm	Outside Diameter mm	Corroded Thickness mm	Factor A	Factor B N./mm ²
10	20	No Calc	581.97	7.595	0.0018126	84.0365
20	30	1297.5	582.605	7.9125	0.0009183	82.6364
30	40	No Calc	...	41.25	No Calc	No Calc
40	50	No Calc	...	41.25	No Calc	No Calc
50	60	700	582.605	7.9125	0.0017823	101.083
60	70	No Calc	...	41.25	No Calc	No Calc
70	80	700	582.605	7.9125	0.0017823	101.083
80	90	No Calc	...	41.25	No Calc	No Calc
90	100	No Calc	...	41.25	No Calc	No Calc

External Pressure Calculations:

From	To	External Actual T. mm	External Required T. mm	External Design Pressure KPa.	External M.A.W.P. KPa.
10	20	10.795	4.74624	103.419	1218.51
20	30	11.1125	5.80324	103.419	1496.32
30	40	44.45	No Calc	103.419	No Calc
40	50	44.45	No Calc	103.419	No Calc
50	60	11.1125	5.21841	103.419	1830.34
60	70	44.45	No Calc	103.419	No Calc
70	80	11.1125	No Calc	...	1830.34
80	90	44.45	No Calc	...	No Calc
90	100	44.45	No Calc	...	No Calc

Minimum

1219

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm	Allowable Length Bet. Stiffeners mm	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	1297.5	37094.2	No Calc	No Calc
30	40	No Calc	No Calc	No Calc	No Calc
40	50	No Calc	No Calc	No Calc	No Calc
50	60	700	14626.4	No Calc	No Calc
60	70	No Calc	No Calc	No Calc	No Calc
70	80	700	No Calc	No Calc	No Calc
80	90	No Calc	No Calc	No Calc	No Calc
90	100	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	47.693	35570.3	35.6759	36656.6	4.7693
20	30	212.657	202088	159.963	208888	21.2657
30	40	107.602	33675	101.385	34236.1	10.7602
40	50	107.602	33675	101.385	34236.1	10.7602
50	60	124.05	122389	93.3116	126355	12.405
60	70	146.815	39129.1	139.608	39690.1	14.6815
70	80	124.05	172676	93.3116	176642	12.405
80	90	146.815	39129.1	139.608	39690.1	14.6815
90	100	222.149	...	222.149	...	22.2149
Total		1239	678330.44	1086	696394.50	123



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm	Y Offset, Dtl. Cent. mm	Z Offset, Dtl. Cent. mm	Description
10	Liqd	34.9204	...	-60.375	...	Liquid: 10
20	Liqd	201.964	...	600	...	Liquid: 20
20	Wght	250.004	...	800	...	flange weight
30	Liqd	36.5262	...	80.8	...	Liquid: 30
40	Liqd	36.5262	...	80.8	...	Liquid: 40
50	Liqd	122.314	...	350	...	Liquid: 50
60	Liqd	36.5262	...	80.8	...	Liquid: 60
70	Liqd	172.57	...	350	...	Liquid: 70
70	Nozl	52.6367	355.196	350	...	N1-Tube Inlet
70	Nozl	52.6367	-355.196	359.481	...	N2-Tube Outlet
80	Liqd	36.5262	...	80.8	...	Liquid: 80
60	FTsh	220.788	...	199.775	...	
60	Tube	518.353	...	-750.225	...	Tubes

Total Weight of Each Detail Type:

Liquid	677.9
Nozzles	105.3
Weights	250.0
Exchanger Components	739.1
Liquid in Tubes	94.8

Sum of the Detail Weights 1867.1 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	1363.4	1363.4	1363.4	1363.4	1363.4	1363.4
Nozzles	105.3	105.3	105.3	105.3	105.3	105.3
Wld Weights	250.0	250.0	250.0	250.0	250.0	250.0
Exchanger	739.1	739.1	739.1	739.1	739.1	739.1
Ope. Liquid	...	...	...	...	...	677.9
Tube Ope Lqd	...	...	...	...	...	94.8



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Element and Detail Weights: Step: 16 10:22am Sep 9, 2026

Test Liquid	...	677.9	...	...	...	...
Tube Tst Lqd	...	94.8	...	...	...	...
Totals	2457.8	3230.5	2457.8	2457.8	2457.8	3230.5

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	2457.8 kg.
Shop Test Wt.	- Fabricated weight + water (full)	3230.5 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	2457.8 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	2457.8 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	2457.8 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	3230.5 kg.
Field Test Wt.	- Empty weight + test medium (full)	3230.5 kg.

Weights used in Vortex Shedding Calculations:

Mass of the Upper 1/3 of the Vertical Vessel (operating)	1006.8 kg.
Mass of the Upper 1/3 of the Vertical Vessel (filled w/water)	1060.9 kg.
Mass of the Upper 1/3 of the Vertical Vessel (empty)	810.2 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	94866.8 cm ³
Weight of Ope Liq in tubes :	94.8 kg.
Weight of Water in tubes :	94.8 kg.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	4654.43	4298.32
20	30	22083.4	21125.8
30	40	5174.52	2491.94
40	50	5174.52	2491.94
50	60	12882	12323.4
60	70	5728.89	2888.92
70	80	12882	12323.4
80	90	5728.89	2888.92
90	100	5939.72	...

Total	80248.3	60832.7 cm ²
Total	86.4	65.5 ft ²

Element and Detail Weights:

From	To	Total Ele. Empty Wgt. kg.	Total. Ele. Oper. Wgt. kg.	Total. Ele. Hydro. Wgt. kg.	Total Dtl. Offset Mom. Kg-m.	Oper. Wgt. No Liquid kg.
10	20	52.4623	87.3827	88.011	...	52.4623
20	30	483.927	685.892	685.892	...	483.927
30	40	118.363	154.889	152.017	...	118.363
40	50	118.363	154.889	152.017	...	118.363
50	60	136.455	258.769	258.769	...	136.455
60	70	382.285	418.811	421.39	...	382.285
70	80	241.728	414.299	414.299	0.16877E-04	241.728
80	90	161.497	198.023	200.602	...	161.497



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Element and Detail Weights:

Step: 16 10:22am

Sep 9, 2026

90	100	244.364	244.364	244.364	...	244.364
----	-----	---------	---------	---------	-----	---------

Cumulative Vessel Weight

From	To	Cumulative Ope Wgt. No Liquid kg.	Cumulative Oper. Wgt. kg.	Cumulative Hydro. Wgt. kg.
10	20	1939.44	2617.32	2617.36
20	30	1886.98	2529.94	2529.35
30	40	1403.05	1844.04	1843.46
40	50	1284.69	1689.16	1691.44
50	60	1166.33	1534.27	1539.42
60	70	1029.87	1275.5	1280.66
70	80	647.589	856.686	859.265
80	90	405.861	442.387	444.966
90	100	244.364	244.364	244.364

The cumulative operating weights no liquid in the column above are the cumulative operating weights minus the operating liquid weight minus any weights absent in the empty condition.

Cumulative Vessel Moment

From	To	Cumulative Empty Mom. Kg-m.	Cumulative Oper. Mom. Kg-m.	Cumulative Hydro. Mom. Kg-m.
10	20	0.16877E-04	0.16877E-04	0.16877E-04
20	30	0.16877E-04	0.16877E-04	0.16877E-04
30	40	0.16877E-04	0.16877E-04	0.16877E-04
40	50	0.16877E-04	0.16877E-04	0.16877E-04
50	60	0.16877E-04	0.16877E-04	0.16877E-04
60	70	0.16877E-04	0.16877E-04	0.16877E-04
70	80	0.16877E-04	0.16877E-04	0.16877E-04
80	90	...	...	...
90	100	...	...	...

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Nozzle Flange MAWP:

Step: 17 10:22am Sep 9,2026

Nozzle Flange MAWP Results: (KPa. & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
N1-Tube Inlet	4781.7	5100.0	82	300	GR 1.1	...	...	...	...
N2-Tube Outlet	4781.7	5100.0	82	300	GR 1.1	...	...	...	...

Shellside Flange Rating**Channelside Flange Rating**

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 4781.667 KPa.

Lowest Flange Pressure Rating was (Amb)[TubeSide]: 5100.005 KPa.

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.

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Center of Gravity Calculation: Step: 18 10:22am Sep 9,2026

Shop/Field Installation Options :

Note : The CG is computed from the first Element From Node

Center of Gravity of the Liquid	1704.540 mm
Center of Gravity of the Nozzles	2825.375 mm
Center of Gravity of the Added Weights (Operating)	850.800 mm
Center of Gravity of the Added Weights (Empty)	850.800 mm
Center of Gravity of the Tubesheet(s)	2449.406 mm
Center of Gravity of the Tubes	1499.406 mm
Center of Gravity of Bare Shell New and Cold	2085.478 mm
Center of Gravity of Bare Shell Corroded	2161.583 mm
Vessel CG in the Operating Condition	1864.368 mm
Vessel CG in the Fabricated (Shop/Empty) Condition	1900.669 mm

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Nozzle Calcs.: N1-Tube Inlet

Nozl: 9 10:22am Sep 9,2026

Input, Nozzle Desc: N1-Tube Inlet

From: 70

Pressure for Reinforcement Calculations	P	905.015	KPa.
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-106 B	
Parent Allowable Stress at Temperature	Sv	118.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	118.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	560.38	mm
Shell Finished (Minimum) Thickness	t	11.1125	mm
Shell Internal Corrosion Allowance	c	3.2000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2820.63	mm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		0.00	deg
Diameter		150.0000	mm
Size and Thickness Basis		Minimum	
Nominal Thickness		40	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.2000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	328.2750	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		80.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



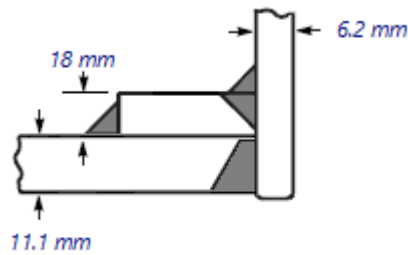
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Nozzle Calcs.: N1-Tube Inlet

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Reinforcement CALCULATION, Description: N1-Tube Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	155.829 mm.
Actual Thickness Used in Calculation	6.223 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 905.01 \cdot 283.39 / (118 \cdot 1.0 - 0.6 \cdot 905.01)$
 $= 2.1837 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 905.01 \cdot 81.11 / (118 \cdot 1.0 - 0.6 \cdot 905.01)$
 $= 0.6250 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	324.4580 mm
Parallel to Vessel Wall, opening length	d	162.2290 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		19.7812 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$

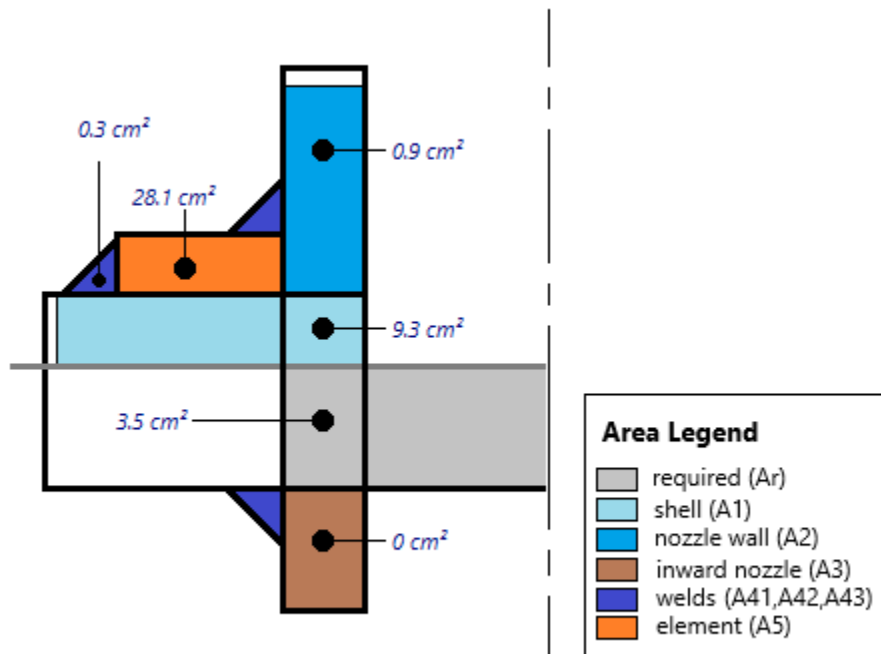


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	3.543	NA	NA
Area in Shell	A1	9.294	NA	NA
Area in Nozzle Wall	A2	0.949	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.325	NA	NA
Area in Element	A5	28.113	NA	NA
TOTAL AREA AVAILABLE	Atot	38.680	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (162.229 \cdot 2.1837 \cdot 1.0 + 2 \cdot 3.023 \cdot 2.1837 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 3.543 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 162.229(1.0 \cdot 7.9125 - 1.0 \cdot 2.184) - 2 \cdot 3.023 \\
 &\quad (1.0 \cdot 7.9125 - 1.0 \cdot 2.1837) \cdot (1 - 1.0) \\
 &= 9.294 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 19.78) \cdot (3.02 - 0.63) \cdot 1.0
 \end{aligned}$$



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Nozzle Calcs.: N1-Tube Inlet

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= 0.949 cm²

Area Available in Welds [A41 + A42 + A43]:

= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Wp²•fr4

= (0.3245) • 1.0 + (0.0) • 1.0 + 0.0² • 1.0

= 0.325 cm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4

= (324.458 - 168.275) • 18.0 • 1.0

= 28.113 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 3.8250 mm

Wall Thickness per UG16(b), tr16b = 5.7000 mm

Wall Thickness, shell/head, internal pressure trb1 = 5.3837 mm

Wall Thickness tb1 = max(trb1, tr16b) = 5.7000 mm

Wall Thickness tb2 = max(trb2, tr16b) = 5.7000 mm

Wall Thickness per table UG-45 tb3 = 9.4200 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]

= min[9.42, max(5.7, 5.7)]

= 5.7000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)

= max(3.825, 5.7)

= 5.7000 mm

Available Nozzle Neck Thickness = 6.2230 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained : 74.7, Allowable : 118.0 N./mm² Passed

Expansion : 0.0, Allowable : 220.3 N./mm² Passed

Occasional : 11.9, Allowable : 156.9 N./mm² Passed

Shear : 49.8, Allowable : 82.6 N./mm² Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 6.223, tr = 0.625, c = 3.2 mm, E* = 1.0

Thickness Ratio = tr * E*/(tg - c) = 0.207, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -8 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 6.223, tr = 0.625, c = 3.2 mm, E* = 1.0

Thickness Ratio = tr * E*/(tg - c) = 0.207, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 6.223, tr = 0.625, c = 3.2 mm, E* = 1.0



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Nozzle Calcs.: N1-Tube Inlet

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Thickness Ratio = $tr * E^* / (tg - c) = 0.207$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $tg = 11.113$, $tr = 2.184$, $c = 3.2$ mm, $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.276$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -26 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(1)(b)), Curve: B

Govrn. thk, $tg = 6.223$, $tr = 0.625$, $c = 3.2$ mm, $E^* = 1.0$
 Thickness Ratio = $tr * E^* / (tg - c) = 0.207$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Gov. MDMT of the Nozzle : -104 °C
 Gov. MDMT of the Reinforcement Pad : -104 °C
 Gov. MDMT of the nozzle to shell joint welded assembly : -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = $905.01 / 5100.00 = 0.177$

Weld Size Calculations, Description: N1-Tube Inlet

Intermediate Calc. for nozzle/shell Welds T_{min} 3.0230 mm
 Intermediate Calc. for pad/shell Welds T_{minPad} 9.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.1161 = 0.7 * t_{min}$	$7.0700 = 0.7 * W_o$ mm
Pad Weld	$4.7500 = 0.5 * T_{minPad}$	$5.6560 = 0.7 * W_p$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A - A1 + 2 \cdot t_n \cdot fr1 \cdot (E1 \cdot t - tr)) \cdot Sv)$
 $= \max(0, (3.5426 - 9.2938 + 2 \cdot 3.023 \cdot 1.0 \cdot (1.0 \cdot 7.9125 - 2.1837)) \cdot 118)$
 $= \max(0, -6503.16)$ Kg

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2 + A5 + A4 - (Wi - Can / .707)^2 \cdot fr2) \cdot Sv$
 $= (0.9487 + 28.1129 + 0.3245 - 0.0 \cdot 1.0) \cdot 118$
 $= 35358.04$ Kg

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot fr1)) \cdot Sv$
 $= (0.9487 + 0.0 + 0.9403 + (0.4784)) \cdot 118$
 $= 2848.54$ Kg

Weld Load [W3]:



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Nozzle Calcs.: N1-Tube Inlet

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$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S \\
 &= (0.9487 + 0.0 + 0.3245 + 28.1129 + (0.4784)) \cdot 118 \\
 &= 35933.65 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 118 \\
 &= 15584. \text{ Kgf}
 \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW} \\
 &= (3.1416/2.0) \cdot 328.275 \cdot 8.0 \cdot 0.49 \cdot 118 \\
 &= 24321. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (T_{hk} - C_{an}) \cdot 0.7 \cdot S_n \\
 &= (3.1416 \cdot 82.626) \cdot (6.223 - 3.2) \cdot 0.7 \cdot 118 \\
 &= 6609. \text{ Kgf}
 \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg} \\
 &= (3.1416/2) \cdot 168.275 \cdot 16.0 \cdot 0.74 \cdot 138 \\
 &= 44039. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng} \\
 &= (3.1416/2.0) \cdot 168.275 \cdot (11.1125 - 3.2) \cdot 0.74 \cdot 118 \\
 &= 18622. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (24321 + 6609) = 30931 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw}) \\
 &= (15584 + 44039 + 18622 + 0) = 78245 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (S_{pew} + T_{ngw} + S_{inw}) \\
 &= (24321 + 18622 + 0) = 42944 \text{ Kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 30930	Kgf, must exceed W = 0 Kgf	or W1 = 35358 Kgf
Path 2-2 = 78244	Kgf, must exceed W = 0 Kgf	or W2 = 2848 Kgf
Path 3-3 = 42943	Kgf, must exceed W = 0 Kgf	or W3 = 35933 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1251 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 12.9311 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 174.0436 mm

Input Echo, WRC297 Item 1, Description: N1-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	3.2000 mm
Shell External Corrosion Allowance	0.0000 mm
Shell Diameter	560.380 mm



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Nozzle Calcs.: N1-Tube Inlet

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Shell Thickness		11.1125	mm
Shell Stress Concentration Factor		1.000	
Vessel Material		SA-106 B	
Vessel Cold S.I. Allowable	Smc	118.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance		3.2000	mm
Nozzle Diameter		155.829	mm
Nozzle Thickness		6.2230	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	328.275	mm
Thickness of Reinforcing Pad	18.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	905.01	KPa.
Radial Load	P	-1147.26	Kgf
Circumferential Shear	Vc	764.84	Kgf
Longitudinal Shear	Vl	764.84	Kgf
Circumferential Moment	Mc	321.11	Kg-m.
Longitudinal Moment	Ml	321.11	Kg-m.
Torsional Moment	Mt	458.73	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{296.346(11.113 - 3.2)}$$

$$= 328.073 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	592.693	mm
Nozzle Outside Diameter	(d) =	168.275	mm
Vessel Thickness used	(T) =	25.912	mm
Nozzle Thickness used	(t) =	3.023	mm
T / t	=	8.572	
d / t	=	55.665	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.358$$

Nr/P	=	0.029
Mr/P	=	0.021
M0/P	=	0.117
N0/P	=	0.369
MrD/Mc	=	0.044
NrDL/Mc	=	0.051



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Nozzle Calcs.: N1-Tube Inlet

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MØd/Mc = 0.289
 NØDL/Mc = 0.367
 MrD/Ml = 0.038
 NrDL/Ml = 0.045
 MØD/Ml = 0.213
 NØDL/Ml = 0.640

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-17	-17	17	17
Outplane Bending (ML)	-35	35	35	-35
Normal Pressure Stress	9	10	9	10

Outplane Stress Summary -26 22 80 -13

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-6	6	6	-6
Inplane Pressure Stress	4	4	4	4

Inplane Stress Summary 0 8 14 -1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary 4 4 2 2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
--	----------------------	---------------------	-------------------------	------------------------

Two * Max Shear Stress 27 24 81 13

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	-10	-10	10	10
Outplane Bending (Mc)	-48	48	48	-48
Outplane Membrane (ML)	0	0	0	0



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Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	4	4	4	4

Outplane Stress Summary	-35	37	81	-38
-------------------------	-----	----	----	-----

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-7	7	7	-7
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	9	10	9	10

Inplane Stress Summary	3	14	20	2
------------------------	---	----	----	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	2	2	4	4
----------------------	---	---	---	---

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	39	37	81	42
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-17	-17	17	17
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	28	29	28	29

Hoop Stress Summary	16	18	52	53
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	7	7	7	7
Axial Bending (P)	47	-47	47	-47
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-49	-49	49	49
Axial Bending (ML)	-190	190	190	-190
Axial Pressure Stress	14	14	14	14



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Axial Stress Summary	-170	114	309	-167
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	9	9	-9	-9
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	33	33	33	33

Shear Stress Summary	42	42	24	24
----------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	206	130	311	226
------------------------	-----	-----	-----	-----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses in the	Left	Left	Right	Right
hoop direction)	Outside	Inside	Outside	Inside

Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-10	-10	10	10
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	28	29	28	29

Hoop Stress Summary	24	25	44	46
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses Normal to	Left	Left	Right	Right
pipe cross-section)	Outside	Inside	Outside	Inside

Axial Membrane (P)	7	7	7	7
Axial Bending (P)	47	-47	47	-47
Axial Membrane (Mc)	-49	-49	49	49
Axial Bending (Mc)	-228	228	228	-228
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	14	14	14	14

Axial Stress Summary	-208	152	347	-205
----------------------	------	-----	-----	------

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress)	Outside	Inside	Outside	Inside

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-9	-9	9	9
Shear due to Torsion	33	33	33	33

Shear Stress Summary	24	24	42	42
----------------------	----	----	----	----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities	Outside	Inside	Outside	Inside

Two * Max Shear Stress	238	157	353	265
------------------------	-----	-----	-----	-----



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Nozzle Calcs.: N1-Tube Inlet

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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{287.346(11.113 - 3.2)}$$

$$= 325.627 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 574.693 mm
 Nozzle Outside Diameter (d) = 328.275 mm
 Vessel Thickness used (T) = 7.912 mm
 Nozzle Thickness used (t) = 83.023 mm
 T / t = 0.095
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.868$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.045
 Mr/P = 0.070
 M0/P = 0.020
 N0/P = 0.030
 MrD/Mc = 0.242
 NrDL/Mc = 0.098
 M0d/Mc = 0.072
 N0DL/Mc = 0.056
 MrD/M1 = 0.052
 NrDL/M1 = 0.048
 M0D/M1 = 0.016
 N0DL/M1 = 0.029

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	21	-21	21	-21
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-14	14	14	-14
Normal Pressure Stress	31	32	31	32
Outplane Stress Summary	39	26	78	6

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	75	-75	75	-75
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-47	47	47	-47
Inplane Pressure Stress	15	15	15	15
Inplane Stress Summary	44	-11	154	-91

Vessel Stresses (N./mm²):



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Nozzle Calcs.: N1-Tube Inlet

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LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)			Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)			1	1	-1	-1
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			3	3	3	3
Shear Stress Summary			5	5	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)			Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress			48	39	154	98

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Membrane (P)			5	5	5	5
Outplane Bending (P)			21	-21	21	-21
Outplane Membrane (Mc)			-8	-8	8	8
Outplane Bending (Mc)			-66	66	66	-66
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			15	15	15	15
Outplane Stress Summary			-32	57	118	-58

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Inplane Membrane (P)			8	8	8	8
Inplane Bending (P)			75	-75	75	-75
Inplane Membrane (Mc)			-14	-14	14	14
Inplane Bending (Mc)			-222	222	222	-222
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			31	32	31	32
Inplane Stress Summary			-121	172	353	-242

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-1	-1	1	1
Torsional Shear (Mt)			3	3	3	3
Shear Stress Summary			1	1	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Two * Max Shear Stress			121	172	353	242



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WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	9	10	9	10	9	10	9	10
Circ. Pl (SUS)	-11	-11	23	23	-1	-1	1	1
Circ. Q (SUS)	-24	24	46	-46	-5	5	9	-9
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-1	-1	1	1	-4	-4	16	16
Long. Q (SUS)	-4	4	8	-8	-37	37	59	-59
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	9.0	10.0	9.0	10.0	9.0	10.0	9.0	10.0
Pm+Pl (SUS)	5.4	4.5	32.0	33.0	8.2	9.2	20.1	20.1
Pm+Pl+Q (Total)	26.6	23.9	78.1	13.4	40.2	37.2	79.3	41.8

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	10.00	118.00	Passed
Pm+Pl (SUS)	33.04	177.00	Passed
Pm+Pl+Q (TOTAL)	79.27	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	31	32	31	32	31	32	31	32
Circ. Pl (SUS)	1	1	9	9	-6	-6	22	22
Circ. Q (SUS)	7	-7	35	-35	-147	147	297	-297
Long. Pm (SUS)	15	15	15	15	15	15	15	15
Long. Pl (SUS)	1	1	15	15	-3	-3	13	13
Long. Q (SUS)	28	-28	122	-122	-45	45	87	-87
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	31.0	32.0	31.0	32.0	31.0	32.0	31.0	32.0
Pm+Pl (SUS)	32.1	33.1	40.1	41.1	25.1	26.1	53.0	54.0
Pm+Pl+Q (Total)	46.2	38.8	152.1	98.1	122.0	173.0	350.1	243.1

Vessel Stress Summation Comparison (N./mm²):



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Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	32.00	118.00	Passed
Pm+Pl (SUS)	54.04	177.00	Passed
Pm+Pl+Q (TOTAL)	350.07	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	28	29	28	29	28	29	28	29
Circ. Pl (SUS)	-11	-11	23	23	-4	-4	16	16
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	14	14	14	14	14	14	14	14
Long. Pl (SUS)	-42	-42	56	56	-42	-42	56	56
Long. Q (SUS)	-143	143	237	-237	-181	181	275	-275
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	9	9	-9	-9	-9	-9	9	9
Shear Q (SUS)	33	33	33	33	33	33	33	33
Pm (SUS)	28.0	29.0	28.0	29.0	28.0	29.0	28.0	29.0
Pm+Pl (SUS)	48.5	49.4	73.6	73.7	55.0	56.0	72.8	72.9
Pm+Pl+Q (Total)	205.9	130.7	309.2	224.2	237.9	157.4	350.8	263.7

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	29.00	118.00	Passed
Pm+Pl (SUS)	73.73	177.00	Passed
Pm+Pl+Q (TOTAL)	350.75	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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Nozzle Calcs.: N2-Tube Outlet

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Input, Nozzle Desc: N2-Tube Outlet

From: 70

Pressure for Reinforcement Calculations	P	904.922	KPa.
Temperature for Internal Pressure	Temp	82	°C
Parent Material		SA-106 B	
Parent Allowable Stress at Temperature	Sv	118.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	118.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	560.38	mm
Shell Finished (Minimum) Thickness	t	11.1125	mm
Shell Internal Corrosion Allowance	c	3.2000	mm
Shell External Corrosion Allowance	co	0.0000	mm
Distance from Bottom/Left Tangent		2830.12	mm
User Entered Minimum Design Metal Temperature		-28.89	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-106 B	
Material UNS Number		K03006	
Material Specification/Type		Smls. pipe	
Allowable Stress at Temperature	Sn	118.00	N./mm ²
Allowable Stress At Ambient	Sna	118.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		180.00	deg
Diameter		150.0000	mm
Size and Thickness Basis		Minimum	
Nominal Thickness		40	
Flange Material		SA-105	
Flange Type		Weld Neck Flange	
Corrosion Allowance	can	3.2000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	150.0000	mm
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	10.0000	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
Pad Material		SA-516 70	
Pad Allowable Stress at Temperature	Sp	138.00	N./mm ²
Pad Allowable Stress At Ambient	Spa	138.00	N./mm ²
Diameter of Pad along vessel surface	Dp	328.2750	mm
Thickness of Pad	te	18.0000	mm
Weld leg size between Pad and Shell	Wp	8.0000	mm
Groove weld depth between Pad and Nozzle	Wgpn	16.0000	mm
Reinforcing Pad Width		80.0000	mm
Flange Class		300	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.



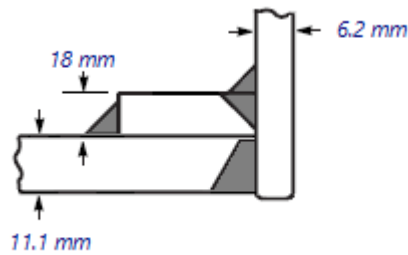
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Nozzle Calcs.: N2-Tube Outlet

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Reinforcement CALCULATION, Description: N2-Tube Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	155.829 mm.
Actual Thickness Used in Calculation	6.223 mm

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]
 $= P \cdot R / (S_v \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 904.92 \cdot 283.39 / (118 \cdot 1.0 - 0.6 \cdot 904.92)$
 $= 2.1835 \text{ mm}$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]
 $= P \cdot R / (S_n \cdot E - 0.6 \cdot P)$ per UG-27 (c)(1)
 $= 904.92 \cdot 81.11 / (118 \cdot 1.0 - 0.6 \cdot 904.92)$
 $= 0.6250 \text{ mm}$

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	324.4580 mm
Parallel to Vessel Wall, opening length	d	162.2290 mm
Normal to Vessel Wall (Thickness Limit), pad side Tlwp		19.7812 mm

The Pad diameter is greater than the Diameter Limit. The excess will not be considered.

Weld Strength Reduction Factor [fr1]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr2]:
 $= \min(1, S_n / S_v)$
 $= \min(1, 118.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr4]:
 $= \min(1, S_p / S_v)$
 $= \min(1, 138.0 / 118.0)$
 $= 1.000$

Weld Strength Reduction Factor [fr3]:
 $= \min(fr2, fr4)$
 $= \min(1.0, 1.0)$
 $= 1.000$



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Nozzle Calcs.: N2-Tube Outlet

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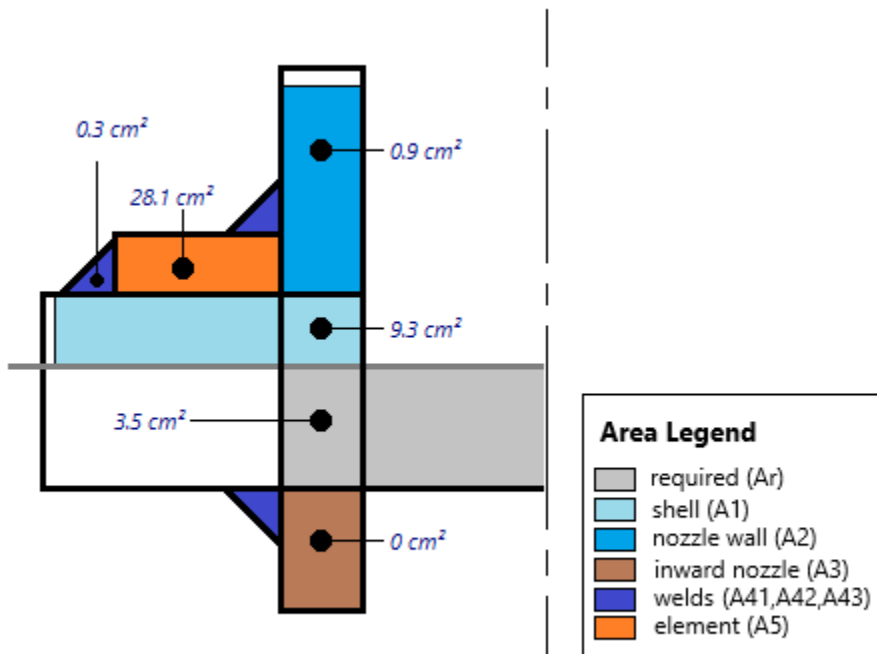


Figure UG 37.1 Nozzle Areas

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	3.542	NA	NA
Area in Shell	A1	9.294	NA	NA
Area in Nozzle Wall	A2	0.949	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	0.325	NA	NA
Area in Element	A5	28.113	NA	NA
TOTAL AREA AVAILABLE	Atot	38.680	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations

90.00 degs.

The area available without a pad is Sufficient.

The area available with the given pad is Sufficient.

Area Required [A]:

$$\begin{aligned}
 &= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)} \\
 &= (162.229 \cdot 2.1835 \cdot 1.0 + 2 \cdot 3.023 \cdot 2.1835 \cdot 1.0 \cdot (1 - 1.0)) \\
 &= 3.542 \text{ cm}^2
 \end{aligned}$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$\begin{aligned}
 &= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1) \\
 &= 162.229(1.0 \cdot 7.9125 - 1.0 \cdot 2.1835) - 2 \cdot 3.023 \\
 &\quad (1.0 \cdot 7.9125 - 1.0 \cdot 2.1835) \cdot (1 - 1.0) \\
 &= 9.294 \text{ cm}^2
 \end{aligned}$$

Area Available in Nozzle Wall Projecting Outward [A2]:

$$\begin{aligned}
 &= (2 \cdot Tlwp) \cdot (tn - trn) \cdot fr2 \\
 &= (2 \cdot 19.78) \cdot (3.02 - 0.62) \cdot 1.0
 \end{aligned}$$



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Nozzle Calcs.: N2-Tube Outlet

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= 0.949 cm²

Area Available in Welds [A41 + A42 + A43]:

= (Wo² - Ar Lost)•Fr3+((Wi-can/0.707)² - Ar Lost)•fr2 + Wp²•fr4

= (0.3245) • 1.0 + (0.0) • 1.0 + 0.0² • 1.0

= 0.325 cm²

Area Available in Element [A5]:

= (min(Dp,DL)-(Nozzle OD))•(min(tp,Tlwp,te)) • fr4

= (324.458 - 168.275) • 18.0 • 1.0

= 28.113 cm²

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures ta = 3.8250 mm

Wall Thickness per UG16(b), tr16b = 5.7000 mm

Wall Thickness, shell/head, internal pressure trb1 = 5.3835 mm

Wall Thickness tb1 = max(trb1, tr16b) = 5.7000 mm

Wall Thickness tb2 = max(trb2, tr16b) = 5.7000 mm

Wall Thickness per table UG-45 tb3 = 9.4200 mm

Determine Nozzle Thickness candidate [tb]:

= min[tb3, max(tb1,tb2)]

= min[9.42, max(5.7, 5.7)]

= 5.7000 mm

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

= max(ta, tb)

= max(3.825, 5.7)

= 5.7000 mm

Available Nozzle Neck Thickness = 6.2230 mm -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	:	74.7,	Allowable	:	118.0 N./mm ²	Passed
Expansion	:	0.0,	Allowable	:	220.3 N./mm ²	Passed
Occasional	:	11.9,	Allowable	:	156.9 N./mm ²	Passed
Shear	:	49.8,	Allowable	:	82.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, tg = 6.223, tr = 0.625, c = 3.2 mm, E* = 1.0

Thickness Ratio = tr * E*/(tg - c) = 0.207, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve A -8 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Min Metal Temp. w/o impact per UG-20(f) -29 °C

Nozzle Neck to Pad Weld for the Nozzle, Curve: B

Govrn. thk, tg = 6.223, tr = 0.625, c = 3.2 mm, E* = 1.0

Thickness Ratio = tr * E*/(tg - c) = 0.207, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -29 °C

Min Metal Temp. at Required thickness (UCS 66.1) -104 °C

Nozzle Neck to Pad Weld for Reinforcement pad, Curve: B

Govrn. thk, tg = 6.223, tr = 0.625, c = 3.2 mm, E* = 1.0



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Nozzle Calcs.: N2-Tube Outlet

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Thickness Ratio = $tr * E^* / (tg - c) = 0.207$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Shell to Pad Weld Junction at Pad OD, Curve: B

Govrn. thk, $tg = 11.113$, $tr = 2.183$, $c = 3.2$ mm, $E^* = 1.0$

Thickness Ratio = $tr * E^* / (tg - c) = 0.276$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-26 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C
Min Metal Temp. w/o impact per UG-20(f)	-29 °C

Nozzle-Shell/Head Weld (UCS-66(a)(b)), Curve: B

Govrn. thk, $tg = 6.223$, $tr = 0.625$, $c = 3.2$ mm, $E^* = 1.0$

Thickness Ratio = $tr * E^* / (tg - c) = 0.207$, Temp. Reduction = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	-29 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-104 °C

Gov. MDMT of the Nozzle	: -104 °C
Gov. MDMT of the Reinforcement Pad	: -104 °C
Gov. MDMT of the nozzle to shell joint welded assembly	: -104 °C

Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c)	-18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b)	-104 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :

Design Pressure/Ambient Rating = $904.92/5100.00 = 0.177$

Weld Size Calculations, Description: N2-Tube Outlet

Intermediate Calc. for nozzle/shell Welds	Tmin	3.0230 mm
Intermediate Calc. for pad/shell Welds	TminPad	9.5000 mm

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	$2.1161 = 0.7 * tmin.$	$7.0700 = 0.7 * Wo$ mm
Pad Weld	$4.7500 = 0.5 * TminPad$	$5.6560 = 0.7 * Wp$ mm

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:

$$= \max(0, (A-A1+2 \cdot tn \cdot fr1 \cdot (E1 \cdot t-tr)) \cdot Sv)$$

$$= \max(0, (3.5422 - 9.2941 + 2 \cdot 3.023 \cdot 1.0 \cdot (1.0 \cdot 7.9125 - 2.1835)) \cdot 118)$$

$$= \max(0, -6504.02) \text{ Kg}$$

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:

$$= (A2+A5+A4-(Wi-Can/.707)^2 \cdot fr2) \cdot Sv$$

$$= (0.9487 + 28.1129 + 0.3245 - 0.0 \cdot 1.0) \cdot 118$$

$$= 35358.07 \text{ Kg}$$

Weld Load [W2]:

$$= (A2 + A3 + A4 + (2 \cdot tn \cdot t \cdot fr1)) \cdot Sv$$

$$= (0.9487 + 0.0 + 0.9403 + (0.4784)) \cdot 118$$

$$= 2848.57 \text{ Kg}$$

Weld Load [W3]:



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$$\begin{aligned}
 &= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f r_1)) \cdot S \\
 &= (0.9487 + 0.0 + 0.3245 + 28.1129 + (0.4784)) \cdot 118 \\
 &= 35933.68 \text{ Kgf}
 \end{aligned}$$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw} \\
 &= (3.1416/2.0) \cdot 168.275 \cdot 10.0 \cdot 0.49 \cdot 118 \\
 &= 15584. \text{ Kgf}
 \end{aligned}$$

Shear, Pad Element Weld [Spew]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_P \cdot W_P \cdot 0.49 \cdot S_{EW} \\
 &= (3.1416/2.0) \cdot 328.275 \cdot 8.0 \cdot 0.49 \cdot 118 \\
 &= 24321. \text{ Kgf}
 \end{aligned}$$

Shear, Nozzle Wall [Snw]:

$$\begin{aligned}
 &= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (T_{hk} - C_{an}) \cdot 0.7 \cdot S_n \\
 &= (3.1416 \cdot 82.626) \cdot (6.223 - 3.2) \cdot 0.7 \cdot 118 \\
 &= 6609. \text{ Kgf}
 \end{aligned}$$

Tension, Pad Groove Weld [Tpgw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot W_{gpn} \cdot 0.74 \cdot S_{eg} \\
 &= (3.1416/2) \cdot 168.275 \cdot 16.0 \cdot 0.74 \cdot 138 \\
 &= 44039. \text{ Kgf}
 \end{aligned}$$

Tension, Shell Groove Weld [Tngw]:

$$\begin{aligned}
 &= (\pi/2) \cdot D_{lo} \cdot (T - c_{as}) \cdot 0.74 \cdot S_{ng} \\
 &= (3.1416/2.0) \cdot 168.275 \cdot (11.1125 - 3.2) \cdot 0.74 \cdot 118 \\
 &= 18622. \text{ Kgf}
 \end{aligned}$$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:

$$= (S_{PEW} + S_{NW}) = (24321 + 6609) = 30931 \text{ Kgf}$$

Failure Path [Path 2-2]:

$$\begin{aligned}
 &= (S_{onw} + T_{pgw} + T_{ngw} + S_{inw}) \\
 &= (15584 + 44039 + 18622 + 0) = 78245 \text{ Kgf}
 \end{aligned}$$

Failure Path [Path 3-3]:

$$\begin{aligned}
 &= (S_{pew} + T_{ngw} + S_{inw}) \\
 &= (24321 + 18622 + 0) = 42944 \text{ Kgf}
 \end{aligned}$$

Summary of Failure Path Calculations:

Path 1-1 = 30930	Kgf, must exceed W = 0 Kgf	or W1 = 35358 Kgf
Path 2-2 = 78244	Kgf, must exceed W = 0 Kgf	or W2 = 2848 Kgf
Path 3-3 = 42943	Kgf, must exceed W = 0 Kgf	or W3 = 35933 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 1251 KPa.

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 12.9311 mm

The Cut Length for this Nozzle is, Drop + Ho + H + T : 174.0436 mm

Input Echo, WRC297 Item 1, Description: N2-Tube Outlet:

Diameter Basis for Cylindrical Shell	ID
Shell Internal Corrosion Allowance	3.2000 mm
Shell External Corrosion Allowance	0.0000 mm
Shell Diameter	560.380 mm



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Nozzle Calcs.: N2-Tube Outlet

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Shell Thickness		11.1125	mm
Shell Stress Concentration Factor		1.000	
Vessel Material		SA-106 B	
Vessel Cold S.I. Allowable	Smc	118.00	N./mm ²
Vessel Hot S.I. Allowable	Smh	118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID		
Nozzle Corrosion Allowance		3.2000	mm
Nozzle Diameter		155.829	mm
Nozzle Thickness		6.2230	mm
Nozzle Stress Concentration Factor		1.000	
Nozzle Material		SA-106 B	
Nozzle Cold S.I. Allowable	SNmc	118.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh	118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter of Reinforcing Pad	328.275	mm
Thickness of Reinforcing Pad	18.0000	mm

Note:

External Forces and Moments in WRC 107/537 Convention:
These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	904.92	KPa.
Radial Load	P	-1147.26	Kgf
Circumferential Shear	Vc	764.84	Kgf
Longitudinal Shear	Vl	764.84	Kgf
Circumferential Moment	Mc	321.11	Kg-m.
Longitudinal Moment	Ml	321.11	Kg-m.
Torsional Moment	Mt	458.73	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - ca)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{296.346(11.113 - 3.2)}$$

$$= 328.073 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter	(D) =	592.693	mm
Nozzle Outside Diameter	(d) =	168.275	mm
Vessel Thickness used	(T) =	25.912	mm
Nozzle Thickness used	(t) =	3.023	mm
T / t	=	8.572	
d / t	=	55.665	

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.358$$

Nr/P	=	0.029
Mr/P	=	0.021
M0/P	=	0.117
N0/P	=	0.369
MrD/Mc	=	0.044
NrDL/Mc	=	0.051



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MØd/Mc = 0.289
 NØDL/Mc = 0.367
 MrD/Ml = 0.038
 NrDL/Ml = 0.045
 MØD/Ml = 0.213
 NØDL/Ml = 0.640

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-17	-17	17	17
Outplane Bending (ML)	-35	35	35	-35
Normal Pressure Stress	9	10	9	10

Outplane Stress Summary -26 22 80 -13

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-1	-1	1	1
Inplane Bending (ML)	-6	6	6	-6
Inplane Pressure Stress	4	4	4	4

Inplane Stress Summary 0 8 14 -1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	3	3	3	3

Shear Stress Summary 4 4 2 2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	27	24	81	13

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	6	6	6	6
Outplane Bending (P)	11	-11	11	-11
Outplane Membrane (Mc)	-10	-10	10	10
Outplane Bending (Mc)	-48	48	48	-48
Outplane Membrane (ML)	0	0	0	0



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Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	4	4	4	4

Outplane Stress Summary	-35	37	81	-38
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stresses parallel to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Inplane Membrane (P)	0	0	0	0
Inplane Bending (P)	2	-2	2	-2
Inplane Membrane (Mc)	-1	-1	1	1
Inplane Bending (Mc)	-7	7	7	-7
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	9	10	9	10

Inplane Stress Summary	3	14	20	2
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Shear stress normal to circumferential plane)	Left Outside	Left Inside	Right Outside	Right Inside

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	3	3	3	3

Shear Stress Summary	2	2	4	4
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
(Stress Intensities)	Left Outside	Left Inside	Right Outside	Right Inside

Two * Max Shear Stress	39	37	81	42
------------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses in the hoop direction)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	-17	-17	17	17
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	28	29	28	29

Hoop Stress Summary	16	18	52	53
---------------------	----	----	----	----

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside

Axial Membrane (P)	7	7	7	7
Axial Bending (P)	47	-47	47	-47
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	-49	-49	49	49
Axial Bending (ML)	-190	190	190	-190
Axial Pressure Stress	14	14	14	14



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Axial Stress Summary	-170	114	309	-167
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
--------------------------------------	---------------	--------------	---------------	--------------

Shear due to (Vc)	9	9	-9	-9
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	33	33	33	33

Shear Stress Summary	42	42	24	24
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
--	---------------	--------------	---------------	--------------

Two * Max Shear Stress	206	130	311	226
------------------------	-----	-----	-----	-----

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Hoop Membrane (P)	6	6	6	6
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	-10	-10	10	10
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	28	29	28	29

Hoop Stress Summary	24	25	44	46
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
---	-----------------------	----------------------	------------------------	-----------------------

Axial Membrane (P)	7	7	7	7
Axial Bending (P)	47	-47	47	-47
Axial Membrane (Mc)	-49	-49	49	49
Axial Bending (Mc)	-228	228	228	-228
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	14	14	14	14

Axial Stress Summary	-208	152	347	-205
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-9	-9	9	9
Shear due to Torsion	33	33	33	33

Shear Stress Summary	24	24	42	42
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Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Outside	Cl Inside	Du Outside	Dl Inside
---	---------------	--------------	---------------	--------------

Two * Max Shear Stress	238	157	353	265
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Stress Computations at the Edge of the Reinforcing Pad:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{R_{\text{mean}}(t - c_a)}$$

$$= 168.275 + 2 \cdot 1.65 \cdot \sqrt{287.346(11.113 - 3.2)}$$

$$= 325.627 \text{ mm}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 574.693 mm
 Nozzle Outside Diameter (d) = 328.275 mm
 Vessel Thickness used (T) = 7.912 mm
 Nozzle Thickness used (t) = 83.023 mm
 T / t = 0.095
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 4.868$$

Note: Re-pad thickness is added to nozzle thickness to simulate the Re-pad.

Nr/P = 0.045
 Mr/P = 0.070
 M0/P = 0.020
 N0/P = 0.030
 MrD/Mc = 0.242
 NrDL/Mc = 0.098
 M0d/Mc = 0.072
 N0DL/Mc = 0.056
 MrD/Ml = 0.052
 NrDL/Ml = 0.048
 M0D/Ml = 0.016
 N0DL/Ml = 0.029

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	5	5	5	5
Outplane Bending (P)	21	-21	21	-21
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	-4	-4	4	4
Outplane Bending (ML)	-14	14	14	-14
Normal Pressure Stress	31	32	31	32
Outplane Stress Summary	39	26	78	6

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	8	8	8	8
Inplane Bending (P)	75	-75	75	-75
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	-7	-7	7	7
Inplane Bending (ML)	-47	47	47	-47
Inplane Pressure Stress	15	15	15	15
Inplane Stress Summary	44	-11	154	-91

Vessel Stresses (N./mm²):



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LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)			Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Outplane Shear (Vc)			1	1	-1	-1
Outplane Shear (Vl)			0	0	0	0
Outplane Shear (Mt)			3	3	3	3
Shear Stress Summary			5	5	1	1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)			Au Top Outside	Al Top Inside	Bu Bottom Outside	B1 Bottom Inside
Two * Max Shear Stress			48	39	154	98

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Membrane (P)			5	5	5	5
Outplane Bending (P)			21	-21	21	-21
Outplane Membrane (Mc)			-8	-8	8	8
Outplane Bending (Mc)			-66	66	66	-66
Outplane Membrane (ML)			0	0	0	0
Outplane Bending (ML)			0	0	0	0
Normal Pressure Stress			15	15	15	15
Outplane Stress Summary			-32	57	118	-58

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Inplane Membrane (P)			8	8	8	8
Inplane Bending (P)			75	-75	75	-75
Inplane Membrane (Mc)			-14	-14	14	14
Inplane Bending (Mc)			-222	222	222	-222
Inplane Membrane (ML)			0	0	0	0
Inplane Bending (ML)			0	0	0	0
Inplane Pressure Stress			31	32	31	32
Inplane Stress Summary			-121	172	353	-242

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Outplane Shear (Vc)			0	0	0	0
Outplane Shear (Vl)			-1	-1	1	1
Torsional Shear (Mt)			3	3	3	3
Shear Stress Summary			1	1	5	5

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)			Cu Left Outside	Cl Left Inside	Du Right Outside	D1 Right Inside
Two * Max Shear Stress			121	172	353	242



WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	9	10	9	10	9	10	9	10
Circ. Pl (SUS)	-11	-11	23	23	-1	-1	1	1
Circ. Q (SUS)	-24	24	46	-46	-5	5	9	-9
Long. Pm (SUS)	4	4	4	4	4	4	4	4
Long. Pl (SUS)	-1	-1	1	1	-4	-4	16	16
Long. Q (SUS)	-4	4	8	-8	-37	37	59	-59
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	9.0	10.0	9.0	10.0	9.0	10.0	9.0	10.0
Pm+Pl (SUS)	5.4	4.5	32.0	33.0	8.2	9.2	20.1	20.1
Pm+Pl+Q (Total)	26.6	23.9	78.1	13.4	40.2	37.2	79.3	41.8

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	10.00	118.00	Passed
Pm+Pl (SUS)	33.04	177.00	Passed
Pm+Pl+Q (TOTAL)	79.27	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Reinforcing Pad Edge (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	31	32	31	32	31	32	31	32
Circ. Pl (SUS)	1	1	9	9	-6	-6	22	22
Circ. Q (SUS)	7	-7	35	-35	-147	147	297	-297
Long. Pm (SUS)	15	15	15	15	15	15	15	15
Long. Pl (SUS)	1	1	15	15	-3	-3	13	13
Long. Q (SUS)	28	-28	122	-122	-45	45	87	-87
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	3	3	3	3	3	3	3	3
Pm (SUS)	31.0	32.0	31.0	32.0	31.0	32.0	31.0	32.0
Pm+Pl (SUS)	32.1	33.1	40.1	41.1	25.1	26.1	53.0	54.0
Pm+Pl+Q (Total)	46.2	38.8	152.1	98.1	122.0	173.0	350.1	243.1

Vessel Stress Summation Comparison (N./mm²):



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Nozzle Calcs.: N2-Tube Outlet

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Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	32.00	118.00	Passed
Pm+Pl (SUS)	54.04	177.00	Passed
Pm+Pl+Q (TOTAL)	350.07	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	28	29	28	29	28	29	28	29
Circ. Pl (SUS)	-11	-11	23	23	-4	-4	16	16
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	14	14	14	14	14	14	14	14
Long. Pl (SUS)	-42	-42	56	56	-42	-42	56	56
Long. Q (SUS)	-143	143	237	-237	-181	181	275	-275
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	9	9	-9	-9	-9	-9	9	9
Shear Q (SUS)	33	33	33	33	33	33	33	33
Pm (SUS)	28.0	29.0	28.0	29.0	28.0	29.0	28.0	29.0
Pm+Pl (SUS)	48.5	49.4	73.6	73.7	55.0	56.0	72.8	72.9
Pm+Pl+Q (Total)	205.9	130.7	309.2	224.2	237.9	157.4	350.8	263.7

Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	29.00	118.00	Passed
Pm+Pl (SUS)	73.73	177.00	Passed
Pm+Pl+Q (TOTAL)	350.75	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

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ASME TS Calc:

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	699.98 KPa.
Shell Maximum Operating Pressure	Psox,max	699.98 KPa.
Shell Minimum Operating Pressure	Psox,min	103.42 KPa.
Shell Thickness	ts	12.7000 mm
Shell Internal Corrosion Allowance	cas	3.2000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	560.380 mm
Shell Temperature for Internal Pressure	Ts	130.00 °C
Shell Material		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K03006
Shell Allowable Stress at Temperature	Ss	118.00 N./mm ²
Shell Allowable Stress at Ambient		118.00 N./mm ²

Channel Data:

Channel Description: Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	899.98 KPa.
Channel Maximum Operating Pressure	Ptox,max	899.98 KPa.
Channel Minimum Operating Pressure	Ptox,min	0.00 KPa.
Channel Thickness	tc	12.7000 mm
Channel Internal Corrosion Allowance	cac	3.2000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	560.380 mm
Channel Design Temperature	TEMPC	82.00 °C
Channel Flange Matl		SA-106 B

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K03006
Channel Allowable Stress at Temperature	Sc	118.00 N./mm ²
Channel Allowable Stress at Ambient		118.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	252
Tube Wall Thickness	et	2.1100 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	1830.00 mm
Straight Tube Length (bet. inner tubsht faces) L		1760.00 mm
Design Temperature of the Tubes		130.00 °C
Tube Material		SA-213 TP316L
Tube Material UNS Number		S31603
Is this a Welded Tube		No
Tube Material Specification used		Smls. tube
Tube Allowable Stress at Temperature		115.00 N./mm ²
Tube Allowable Stress At Ambient		115.00 N./mm ²
Tube Yield Stress At design Temperature	Syt	135.80 N./mm ²
Tube Pitch (Center to Center Spacing)	P	28.5750 mm
Tube Layout Pattern		Triangular
Fillet Weld Leg	af	1.6510 mm
Groove Weld Leg	ag	1.6510 mm
Tube-Tubesheet Joint Weld Type		Seal/No Weld
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	222.201 mm
Largest Center-to-Center Tube Distance	Ul	0.0000 mm



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ASME TS Calc:

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Length of Expanded Portion of Tube	ltx	46.8000 mm
Tube-side pass partition groove depth	hg	7.0000 mm
Area of Tube Layout	Ap	1127.2 cm ²
Perimeter of Tube Layout	Cp	1307.9 mm

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral with Channel, Conf. e

Tubesheet Design Metal Temperature	T	130.00 °C
Tubesheet Material		SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	70.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	3.2000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	3.2000 mm
Tubesheet Outside Diameter	A	750.000 mm

Additional Data for Stepped Tubesheets:

Is the Tubesheet Stepped?		No
Area of the Untubed Lanes	AL	177.2 cm ²
Junction Stress Reduction option		Perform Plastic Calculation
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Shell Side Min. Design Pressure	Psd,min	103.4190 KPa.

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Shell
Flange Outside Diameter	A	750.000 mm
Flange Inside Diameter	B	536.575 mm
Flange Face Outside Diameter	Fod	641.350 mm
Flange Face Inside Diameter	Fid	536.575 mm
Gasket Outside Diameter	Go	641.350 mm
Gasket Inside Diameter	Gi	536.575 mm
Small end Hub thk.	g0	11.1125 mm
Large end Hub thk.	g1	36.5125 mm
Gasket Factor,	m	3.00
Gasket Design Seating Stress	y	68.95 N./mm ²
Flange Facing Sketch		Code Sketch 1a
Gasket Thickness	tg	3.1750 mm
Full face Gasket Flange Option		Program Selects

Bolting Information:

Diameter of Bolt Circle	C	692.150 mm
Nominal Bolt Diameter	dB	31.7500 mm
Thread Series		TEMA Thread Series
Number of Bolts	n	20
Bolt Material		SA-193 B7
Bolt Allowable Stress At Temperature	Sb	172.00 N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00 N./mm ²
Weld between Flange and Shell/Channel		0.0000 mm

Note: The full bolt load Sa*Ab, was used for all load cases.

Tubesheet Integral with		Channel
Tubesheet Extended as Flange		Yes
Thickness of Extended Portion of Tubesheet	Tf	54.0000 mm
Is Bolt Load Transferred to the Tubesheet		Yes

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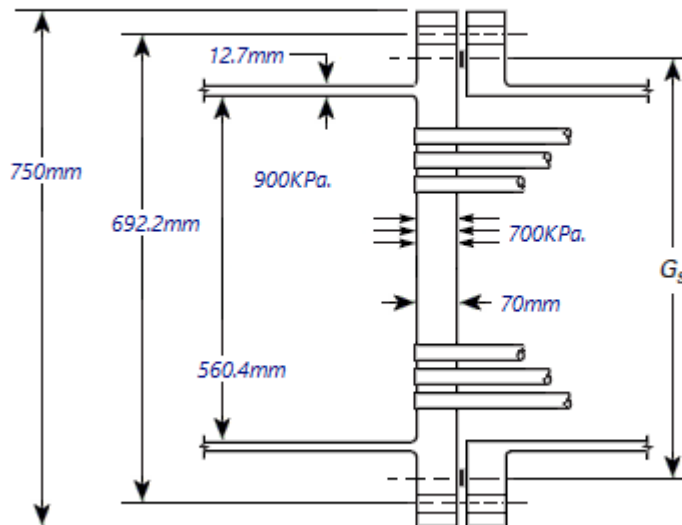
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ASME TS Calc:

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Tubesheet Gasketed w/Shell and Integral w/Channel, Extended as a Flange



Configuration e:

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with $C \leq 0.3\%$

Elastic Mod. at Design Temperature	130.0 °C	0.19629E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Carbon Steels with $C \leq 0.3\%$

Elastic Mod. at Design Temperature	82.0 °C	0.19921E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TM-1 Material Group G

Elastic Mod. at Tubsht. Design Temp.	130.0 °C	0.18733E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.19512E+09 KPa.

TubeSheet - TM-1 Carbon Steels with $C \leq 0.3\%$

Elastic Mod. at Design Temperature	130.0 °C	0.19629E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:

$$= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P) \text{ per Appendix 1-1 (a)(1)}$$

$$= (1003.4 \cdot (19.05/2 - 0)) / (115.0 \cdot 1.0 + 0.4 \cdot 1003.4)$$

$$= 0.0828 + 0.0000 = 0.0828 \text{ mm}$$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart HA-4 at 130.00 °C
Elastic Modulus for Material 183893936.00 KPa.



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ASME TS Calc:

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Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.1100	19.05	1982.20	9.03	50.0000	0.0134948	81.13

EMAWP = $(2.167 / (D/T) - 0.0833) * B = 12714.3848 \text{ KPa.}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
0.3298	19.05	1982.20	57.76	50.0000	0.0003298	30.32

EMAWP = $(4 * B) / (3 * (D/T)) = (4 * 30.3232) / (3 * 57.7551) = 700.0015 \text{ KPa.}$

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	0.3298 mm
Allowable Internal Pressure at Corroded thickness	27949.72 KPa.
Required Internal Design Pressure	1003.40 KPa.
Allowable External Pressure at Corroded thickness	12714.38 KPa.
Required External Design Pressure	699.98 KPa.
Required Thickness due to Shell Side pressure	0.3298 mm

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Intermediate Calculations For Tubesheets Extended As Flanges:

ASME Code, Section VIII Division 1, 2025

Gasket Contact Width,	$N = (Goc - Gic) / 2$	52.388 mm
Basic Gasket Width,	$b0 = N / 2$	26.194 mm
Effective Gasket Width,	$b = \text{sqrt}(b0) * 2.5$	12.892 mm
Gasket Reaction Diameter,	$G = Go - 2.0 * b$	615.566 mm

Results for ASME U-tube Tubesheet Calculations for Configuration e:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$= 2 * ro + dt$
 $= 2 * 222.201 + 19.05$
 $= 463.452 \text{ mm}$

Basic ligament efficiency for shear [μ]:

$= (p - dt) / p$
 $= (28.575 - 19.05) / 28.575$
 $= 0.333$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$= Gs / Do$ (Configurations d, e, f)
 $= 615.5656 / 463.452$
 $= 1.3282$

Ratio [pc]:

$= Dc / Do$ (Configurations a, e, f)
 $= 560.38 / 463.452$
 $= 1.2091$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$= ltx / h$
 $= 46.8 / 70.0$



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ASME TS Calc:

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$$= 0.669 \text{ (must be } 0 \leq \rho \leq 1 \text{)}$$

Effective tube hole Diameter [d*]:

$$\begin{aligned} &= \max(dt - 2tt(EtT/E) \cdot (StT/S) \cdot (\rho); dt - 2tt) \\ &= \max(19.05 - 2 \cdot 2.11 (0.18733E+09/1.9629E+09) \cdot \\ &\quad (114/137) \cdot (0.669); 19.05 - 2 \cdot 2.11) \\ &= 16.8062 \text{ mm} \end{aligned}$$

Effective tube pitch [p*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(A1; 4 \cdot Do \cdot p) / (\pi \cdot Do^2)) } \\ &= 28.575 / \sqrt{ (1 - 4 \cdot \min(177.16 ; 4 \cdot 463.452 \cdot 28.575) / (\pi \cdot 463.452^2)) } \\ &= 30.2050 \text{ mm} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (30.205 - 16.8062) / 30.205 \\ &= 0.4436 \end{aligned}$$

$$h/p = 2.449694 ; \mu^* = 0.443595 ; E^* = 92387784 \text{ KPa.}$$

$$E^*/E = 0.470670 ; \nu^* = 0.303679$$

Note: As h/p (2.450) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Intermediate parameters for Tubesheet Gasketed on the Shell Side:

β_s , k_s , λ_s , δ_s and ω_s are all = 0

Channel coefficient [β_c]:

$$\begin{aligned} &= ((12(1 - \nu_c^2))^{1/4}) / \sqrt{ ((Dc + tc) tc) } \\ &= ((12(1 - 0.3^2))^{1/4}) / \sqrt{ ((560.38 + 12.7) 12.7) } \\ &= 0.0213 \text{ 1/mm} \end{aligned}$$

Channel coefficient [k_c]:

$$\begin{aligned} &= \beta_c \cdot E_c \cdot tc^3 / (6(1 - \nu_s^2)) \\ &= 0.021 \cdot 0.19921E+09 \cdot 12.7^3 / (6(1 - 0.3^2)) \\ &= 1592519296.0000 \text{ KPa} \cdot \text{mm}^2 \end{aligned}$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$\begin{aligned} &= 6(Dc + tc/\cos(\theta)) \cdot k_c/h^3 \cdot (CmTheta + \\ &\quad h \cdot \beta_c \cdot (Cm\Delta + CqTheta)/2 + (h \cdot \beta_c)^2/2 \cdot Cq\Delta) \\ &= 6(560.38 + 12.7/\cos(0)) \cdot 0.15925E+1/70.0^3 \cdot (1.0 + \\ &\quad 70.0 \cdot 0.021 \cdot (1.0 + 1.0)/2 + (70.0 \cdot 0.021)^2/2 \cdot 1.0) \\ &= 57535732.0 \text{ KPa.} \end{aligned}$$

Channel coefficient [δ_c]:

$$\begin{aligned} &= Dc^2 / (4 \cdot E_c \cdot tc) \cdot (1 - Dc \cdot \nu_c / (2(Dc + tc))) \\ &= 560.38^2 / (4 \cdot 0.19921E+09 \cdot 12.7) \cdot (1 - 560.38 \cdot 0.3 / (2(560.38 + 12.7))) \\ &= 0.26477E-01 \text{ mm/N./mm}^2 \end{aligned}$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ω_c]:

$$\begin{aligned} &= p_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + Cq\Delta \cdot h \cdot \beta_c^2) - \\ &\quad \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)) \\ &= 1.209 \cdot 0.15925E+1 [0.026477 (1.0 + 0.021 \cdot 1.0 \cdot 70.0) - \\ &\quad 0 (1.0/70.0 + 1.0 \cdot 0.021/2)) \\ &= 2706.9338 \text{ mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:



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ASME TS Calc:

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Diameter ratio [K]:

$$= A / D_o$$

$$= 750.0/463.452$$

$$= 1.6183$$

Coefficient [F]:

$$= (1 - \nu^*) / E^* \cdot (\lambda c + E \cdot \ln(K))$$

$$= (1 - 0.3) / 92387784 \cdot (57535732 + 0.19629E+09 \cdot \ln(1.62))$$

$$= 1.1458$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= D_o^2/16 \cdot [(p_s - 1) \cdot (p_s^2 + 1) \cdot P_s - (p_t - 1) \cdot (p_t^2 + 1) \cdot P_t]$$

$$= 463.452^2/16 \cdot [(1.328 - 1) \cdot (1.328^2 + 1) \cdot 699.984 - (1.209 - 1) \cdot (1.209^2 + 1) \cdot 899.979]$$

$$= 2304219.5000 \text{ KPa} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]

$$= M_{ts} + (\omega c \cdot P_t) + W^* \cdot (C - G_s) / (2 \cdot \pi \cdot D_o)$$

$$= 2304219 + (2436183) + 210235 \cdot (692.15 - 615.566) / (2 \cdot \pi \cdot 463.452)$$

$$= 58961012.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^*) - D_o^2/32 \cdot F(P_s - P_t) / (1 + F)$$

$$= (58961012) - 463.452^2/32 \cdot 1.146 (699.98 - 899.98) / (1 + 1.15)$$

$$= 28194262.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= M_p + D_o^2/64 \cdot (3 + \nu^*) \cdot (P_s - P_t)$$

$$= 28194262 + 463.452^2/64 \cdot (3 + 0.304) \cdot (699.98 - 899.98)$$

$$= 25976848.0000 \text{ KPa} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|M_p|; |M_o|)$$

$$= \max(|28194262|; |25976848|)$$

$$= 28194262.0000 \text{ KPa} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 28194262 / (0.4436 (70.0 - 7.0)^2)$$

$$= 96.0880 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot S$$

$$= 2 \cdot 138.0$$

$$= 275.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu^* (h - h_g')^2)$$

$$= 6 \cdot 19459410 / (0.4804 (36.7217 - 7.0)^2)$$

$$= 275.1297 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + C_{ats} + C_{atc}$$

$$= 36.7217 + 0 + 0$$

$$= 36.7217 \text{ mm}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS})$$

$$= \max(36.7217; 0.4684)$$



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ASME TS Calc:

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= 36.7217 mm

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [| Ps - Pt |]:

= | Ps - Pt |

= | 699.98 - 899.98 |

= 199.995 KPa.

Shear Stress check [τ limit]:

= $4 \cdot \mu \cdot h/Do \cdot \min(0.8 \cdot S; 0.533 \cdot Sy)$

= $4 \cdot 0.333 \cdot 70.0/463.45 \cdot \min(0.8 \cdot 137.997; 0.533 \cdot 222.198)$

= 22.233 N./mm²

| Ps - Pt | is $\leq$ the Shear Stress limit (τ limit). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

= $1/(4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot Sy)) \cdot (4 \cdot Ap/Cp) \cdot | Ps - Pt | + Cats + Catc$

= $1/(4 \cdot 0.333 \cdot 110.398) \cdot (4 \cdot 1127.2/1307.93) \cdot | 699.98 - 899.98 | + 0$

= 0.4684 mm

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Channel thickness of: 12.700 mm

Min. Channel len. adjacent to the tubesheet is: 151.850mm

Axial Channel Membrane Stress [$\sigma_{c,m}$]:

= $Dc^2 / (4 \cdot tc(Dc + tc) \cdot \cos(\theta)) \cdot Pt$

= $560.38^2 / (4 \cdot 12.7(560.38 + 12.7) \cdot \cos(0)) \cdot 899.979$

= 9.708 N./mm²

Axial Channel Bending Stress [$\sigma_{c,b}$]:

= $6 \cdot kc/tc^2 \cdot [(\delta_c \cdot Cm\Delta \cdot \beta_c - \zeta_c \cdot Cm\Theta) \cdot Pt -$

$6((1 - \nu^*)/E^*) \cdot Do/h^3 \cdot (Cm\Theta +$

$h \cdot \beta_c/2 \cdot Cm\Delta) \cdot (Mp + Do^2/32 \cdot (Ps - Pt))]$

= $6 \cdot 358020.75/12.7^2 \cdot [(0.026477 \cdot 1.0 \cdot 0.021 - 0 \cdot 1.0) \cdot 899.979 -$

$6((1 - 0.304)/92387784) \cdot 463.452/70.0^3 \cdot (1.0 +$

$0.021 \cdot 70.0/2 \cdot 1.0) \cdot (28194262 + 463.452^2/32 \cdot (699.984 - 899.979))]$

= -139.614 N./mm²

Channel Membrane + Bending Stress [σ_c]:

= | $\sigma_{c,m}$ | + | $\sigma_{c,b}$ | (Should be $\leq 1.5 \cdot Sc$)

= 149.323 N./mm² (Should be ≤ 177.00)

Tube Metal Area [At]:

= $\pi(do - t)t$

= $\pi(19.05 - 2.11)2.11$

= 1.123 cm²

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	96.1 <=	276.0 N./mm ²	Ok
Tubesheet shear stress	0.7 <=	110.4 N./mm ²	Ok
Stress in Channel at Tubesheet	149.3 <=	177.0 N./mm ²	Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm)	Required	Actual	P/F
Tubesheet Thickness :	36.722	70.000	Ok



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ASME TS Calc:

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Tube-Tubesheet Fillet Weld Leg :	0.000	1.651	Ok
Tube-Tubesheet Groove Weld Leg :	0.000	1.651	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 8 Load Cases:

Case#	--Reqd. Thk. + CA		---- Tubesheet		Stresses		Case Type	Pass/Fail
	Tbsht	Extnsn	Bend	Allwd	Shear	Allwd		
D1uc	24.175	41.995	34	276	4	110	Ps+Pt D1	Ok
D2uc	43.008	...	119	276	3	110	Ps+Pt D2	Ok
D3uc	36.722	...	96	276	1	110	Ps+Pt D3	Ok
D4uc	33.765	...	85	276	...	110	Ps+Pt D4	Ok
D1c	28.591	41.995	40	276	4	110	Ps+Pt-c D1	Ok
D2c	50.020	...	141	276	3	110	Ps+Pt-c D2	Ok
D3c	43.607	...	111	276	1	110	Ps+Pt-c D3	Ok
D4c	41.611	...	101	276	...	110	Ps+Pt-c D4	Ok
Max:	50.0197	41.995 mm		0.509		0.037 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:

Shell-side and Tube-side Design or Operating Pressures derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the Load Case Tables

[c]:

With or Without Corrosion Allowance

[D1, D2, D3]:

Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:

Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Req'd. Thk.		Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
	Shell	Channel					
D1uc	...	...	...	...	74	177	Ok
D2uc	...	...	...	...	199	447	Ok
D3uc	...	...	...	...	149	177	Ok
D4uc	...	...	...	...	154	177	Ok
D1c	...	...	...	...	101	177	Ok
D2c	...	...	...	...	247	447	Ok
D3c	...	...	...	...	179	447	Ok
D4c	...	...	...	...	191	447	Ok
Max	...	...	...	...		0.870	

Summary of Thickness Comparisons for 8 Load Cases:

Thickness (mm)	Required	Actual	P/F
Tubesheet thickness :	50.020	70.000	Ok
Tubesheet thickness flanged extension :	41.995	54.000	Ok
Tube thickness :	0.330	2.110	Ok
Tube-tubesheet fillet weld leg :	0.000	1.651	Ok
Tube-tubesheet groove weld leg :	0.000	1.651	Ok

Given Shell Thickness: 12.7000 mm

Given Channel Thickness: 12.7000 mm



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ASME TS Calc:

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Min Channel length of thk, (12.700) adj. to tubesheet: 151.850 mm

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	7112.54	1.000	3597.07	0.998
Tubesheet Shear Stress	27156.54	1.000	27156.54	1.000
Tubesheet-Channel Junction	4294.07	1.000	3597.07	0.998
Tube Pressure Stress	27949.71	1.000	12714.38	1.000
Tubesheet Extension Stress	No Calc	No Calc	4610.11	...
Minimum MAWP	4294.07		3597.07	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	8552.52	1.000	4115.99	0.997
Tubesheet Shear Stress	29889.28	1.000	29889.28	1.000
Tubesheet-Channel Junction	6318.39	1.000	4115.99	0.997
Tube Pressure Stress	27949.71	1.000	15530.96	1.000
Tubesheet Extension Stress	No Calc	No Calc	4610.11	...
Minimum MAPnc	6318.39		4115.99	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Channel Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):
 = max(tubesheet thk/4, min(tubesheet thk, channel thickness))
 = max(70.0/4, min(70.0, 12.7))
 = 17.500 mm

Thickness Ratio = 0.17, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

Shell Side MDMT calculation:

Governing thickness for the shell side:
 = tubesheet thickness/4
 = 70.0/4
 = 17.500 mm

Thickness Ratio = 0.17, Temperature Reduction per Fig. UCS 66.1 = 78 °C

Min Metal Temp. w/o impact per UCS-66, Curve B -11 °C
 Min Metal Temp. at Required thickness (UCS 66.1) -104 °C
 Min Metal Temp. w/o impact per UG-20(f) -29 °C

where the MDMT reduction ratio per UCS 66 (b)(1)(b) is:
 = max(pt/Tubeside MAPnc, ps/Shellside MAPnc), must be ≤ 1
 = max(899.98/6318.39, 699.98/4115.99)
 = 0.170



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ASME TS Calc:

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PV Elite® Pressure Vessel Design Notes

Project Information	
Project Number	
Client	
Date	
Designer	
Design Code	
Additional Standards	

Design Conditions		
Internal Design Pressure:		
External Design Pressure:		
Design Int. Temperature:		
Design Ext. Temperature:		
Corrosion Allowance:		

Vessel Geometry		
Vessel Type		
Diameter		
Tangent-to-Tangent Length		
Head Type		
Number of Segments		

Material Specifications	
Shell:	
Head:	
Nozzle:	

Nozzle/Opening Details	ID/OD	Size	Rating	Matl.	Reinf.

Miscellaneous		
Support Type		
Number of Supports		
Support Material		
Seismic Code		
Wind Code		
Insulation		
Pressure Test		
Radiography		
Fatigue Analysis		

Analysis Notes: